

IDENTIFICATION

PRODUCT CODE: MAINDEC-8E-D6AB-D
PRODUCT NAME: PDP8-E XY8-E PLOTTER
CONTROL AND DISPLAY
DIAGNOSTIC PROGRAM
DATE CREATED: MAY 18, 1971
MAINTAINER: DIAGNOSTIC PROGRAMMING
GROUP
AUTHOR: LEONARD E. BEYERSDORFER

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1.

ABSTRACT

THE XY8-E PLOTTER OPTION CONTROL AND DISPLAY DIAGNOSTIC PROGRAM TESTS THE OVERALL OPERATION OF THE XY8-E CONTROL MODULE AND THE CALCOMP PLOTTER (SERIES 500 THROUGH 700), HOUSTON DP10 OR EDP10, OR EQUIVALENT, INSTALLED WITHIN A PDP8-E SYSTEM.

THIS PROGRAM IS SET UP TO USE DEVICE CODE 50. IF THE DEVICE CODE IN THE SYSTEM UNDER TEST IS OTHER THAN 50, LOAD ADDRESS 0225, CLEAR ALL SWITCHES, SET SR3-8 TO NEW DEVICE CODE, THEN DEPRESS CLEAR FOLLOWED BY CONTINUE. THE PROGRAM WILL HALT WITH THE MA=0227 AND THE DEVICE CODE IN THE AC.

ALL TIMING CHECKED IN THIS PROGRAM IS BASED UPON A 72.7 MILLISEC FLAG SETTING TIME FOR PEN UP AND PEN DOWN MOVEMENTS, AND A 7.5 MILLISEC FLAG SETTING TIME FOR ALL OTHER MOVEMENTS. IF THE SYSTEM IS CONFIGURED DIFFERENTLY, CHANGE THE CONTENTS OF THE FOLLOWING LOCATIONS AS SHOWN BELOW.

(A = PEN UP AND PEN DOWN FLAG SETTING TIME.)

(B = FLAG SETTING TIME FOR ALL OTHER MOVEMENTS.)

RELATIVE	ABSOLUTE	OLD	NEW
K70MIN	0115	0062	70% OF A
K70MAX	0116	0055	60% OF A
K5MIN	0117	0005	70% OF B
K5MAX	0120	0005	60% OF B
K5ALL	0121	0010	130% OF B

(1.2 CONT'D)

VECTOR TYPES:*

8 VECTOR
24 VECTOR

*PATTERNS 4 AND 5 DIFFER FOR 8 AND 24 VECTOR PLOTTERS.

INCREMENT SIZES:

0.1 MM (8 VECTOR ONLY)

.00125 INCH
.0025 INCH
.005 INCH
.01 INCH

2. REQUIREMENTS

2.1 EQUIPMENT

PDP8-E WITH XY8-E OPTION, CALCOMP OR HOUSTON PLOTTER WITH SPECIFICATIONS
FITTING THOSE DELINEATED IN PARAGRAPH 1.2 ABOVE (FOR DISPLAY
TEST ONLY), AND TELETYPE.

2.2 STORAGE

4K OF CORE REQUIRED (FIELD 0).

3. LOADING PROCEDURE

LOAD THE PROGRAM USING THE BIN LOADER.

4.2 STARTING ADDRESSES (OPTIONAL)

STARTING ADDRESS	RESULT
200	RUN COMPLETE TEST.
201	RUN COMPLETE TEST WITHOUT INIT AND TEST 0,
202	INITIALIZE ONLY.
204	DISPLAY TEST ONLY.
205	TEST 0, THEN HALT.
210	TEST 1, THEN HALT.
212	TEST 2, THEN HALT.
214	TEST 3, THEN HALT.
216	TEST 4, THEN HALT.
220	TEST 5, THEN HALT.
222	TEST 6, THEN HALT.
225	CHANGE DEVICE CODE (REFER TO PARAGRAPH 1)

5. OPERATING PROCEDURE

5.1 CONTROL TEST

5.1.1 SWITCH REGISTER OPTIONS

SR BIT SET	YIELD
0	INHIBIT ERROR HALTS.
1	INHIBIT ERROR PRINTOUTS.
2	BELL ON ERROR.
3	LOOP 1.
4	LOOP 2.
5	LOOP 3.
9	INHIBIT RUNNING TEST0.
*10	TAKE ERROR CONTINUE EXIT.
11	LOOP ON CONTROL TEST (EXCEPT TEST0).

*SETTING SR10 MAY RESULT IN MISLEADING ERROR PRINTOUTS OCCURRING AFTER THE FIRST ERROR PRINTOUT. THIS OPTION SHOULD BE USED ONLY AFTER THE PROGRAM LISTING HAS BEEN CONSULTED TO DETERMINE THE CONSEQUENCES. (ERROR 6E IS NOT AFFECTED BY THIS OPTION.)

5.2

DISPLAY TEST

CAUTION

PRIOR TO RUNNING THE CONTROL AND DISPLAY TESTS, ENSURE THAT ALL PLOTTER MANUAL COMMAND SWITCHES ARE IN THEIR NEUTRAL POSITION.

5.2.1 MONITOR COMMANDS

THE DISPLAY TEST IS CONTROLLED BY A MONITOR WHICH RESPONDS TO THE FOLLOWING COMMANDS FROM THE TTY. TO RUN COMPLETE DISPLAY TEST, POSITION PEN TO A POINT AT LEAST 2 INCHES FROM ANY PHYSICAL STOP AND TYPE "CD-".

NOTE: "-" INDICATES RETURN KEY.

"L" INDICATES ALTMODE.

"N" INDICATES ANY OCTAL NUMBER.

TYPING RUBOUT ABORTS THE COMPLETE COMMAND STRING.

COMMAND	RESULT
"CD-"	DRAW COMPLETE DISPLAY TEST
"CD;PNN-"	DRAW PATTERN NN ONCE - LIMITED TO ONE PATTERN
"CD;PNNLNN;PNNLNN-"	DRAW PATTERN NN LINE NN FOLLOWED BY PATTERN NN LINE NN - LIMITED TO 10 PATTERN/LINE COMBINATIONS.
"CL-"	LOOP ON COMPLETE DISPLAY TEST.
"CL;PNN-"	LOOP ON DRAWING PATTERN NN - LIMITED TO ONE PATTERN.
"CL;PNNLNN;PNNLNN-"	LOOP ON DRAWING PATTERN NN LINE NN FOLLOWED BY PATTERN NN LINE NN - LIMITED TO 10 PATTERN/LINE COMBINATIONS.
*"C-"	CONTINUE EXECUTING PREVIOUS COMMAND.
*"CL-"	CONTINUE EXECUTING PREVIOUS COMMAND AND UNCONDITIONALLY LOOP.
*"CD-"	CONTINUE EXECUTING PREVIOUS COMMAND AND STOP LOOPING.
"CT-"	TERMINATE DISPLAY TEST MONITOR AND RETURN TO CONTROL TEST EXECUTIVE STARTING AT TEST1.
"I-"	TERMINATE DISPLAY TEST MONITOR, GO TO INIT FOR NEW SPECIFICATION INPUT, AND RETURN TO DISPLAY MONITOR WHEN COMPLETED.

6. ERRORS

6.1 CONTROL TEST ERRORS

6.1.1 ERROR HALTS

ONLY TWO ERROR HALTS BEAR EXPLANATION SINCE THEY HAVE NO ACCOMPANYING TYPEOUTS.

PC ERROR

1012 THE SIGNAL "INITIALIZE" DOES NOT CLEAR THE PLOTTER FLAG, OR PLSF SKIPS WHEN FLAG IS CLEAR, DEPRESS START TO RETEST.

1027 THE SIGNAL "INITIALIZE" DOES NOT SET INTERRUPT ENABLE, OR FLAG SET AND INTERRUPT ENABLE SET DO NOT RESULT IN AN INTERRUPT REQUEST. DEPRESS START TO RETEST.

6.1.2 ERROR HANGUP.

MA ERROR

1014/1015 PLLR WITH AC=00 DOES NOT SET PLOTTER FLAG OR PLSF DOES NOT SKIP WHEN FLAG IS SET. TO RECOVER, INHIBIT TEST0 AND START FROM 200.

6.1.3 ERROR PRINTOUTS.

DURING THE CONTROL TEST, IF AN ERROR OCCURS A GENERAL ERROR HEADING IS DISPLAYED AND SPECIFIC ERROR INFORMATION IS INSERTED UNDER THE APPLICABLE HEADING. THE HEADING AND RELATED DEFINITIONS ARE AS FOLLOWS:

"ERROR" ERROR CODE FILL IN.

"PC" ADDRESS AT WHICH THE ERROR WAS DETECTED.

"DR" THE DATA WHICH SHOULD HAVE BEEN LOADED INTO THE DIRECTION REGISTER AT THE TIME THE ERROR OCCURRED. (THIS IS DISPLAYED ONLY IF APPLICABLE TO THE ERROR.)

REFER TO TABLE 1 FOR CONTROL TEST ERROR PRINTOUT INTERPRETATION AND RECOMMENDED RECOVERY.

7. NORMAL PRINTOUTS

ALL NORMAL PRINTOUTS ARE SELF-EXPLANATORY.

8. RESTRICTIONS

CAUTION

PRIOR TO RUNNING THE CONTROL AND DISPLAY TESTS, ENSURE
THAT ALL PLOTTER MANUAL COMMAND SWITCHES ARE IN THEIR
NEUTRAL POSITION.

CAUTION

IF THE "CM-" COMMAND OPTION IS UTILIZED, ENSURE THAT NO
CONFLICTING COMMANDS ARE ENTERED INTO THE DIRECTION
REGISTER VIA THE SWITCH REGISTER.

KEY TO TABLE 1 (CONTROL TEST ERROR RECOVERY TABLE)

ERROR: NX N INDICATES THE TEST WHICH FIRST CHECKS FOR THIS ERROR CONDITION (PRIMARY TEST). X INDICATES THE APPROXIMATE LOGICAL SEQUENCE WITHIN THAT TEST (IN MOST CASES).

PC: NNNN ADDRESS AT WHICH PROGRAM BRANCHED TO ERROR ROUTINE.

DR: NN CONTENTS OF AC (AND THEREFORE SHOULD BE CONTENTS OF DR) AFTER PLLR OR CFLR IOT'S.

TST: N TEST IN WHICH ERROR WAS DETECTED.

FAILURE: GENERAL ERROR DESCRIPTION.

RECOVERY: LOOP N (SCOPING PRIOR TO LOOPING MAY BE BENEFICIAL IN SOME CASES.) SET LOOP N SR BIT AND DEPRESS CONTINUE, SR BITS 0, 1 AND 2 MAY BE SET IN ANY CONFIGURATION DESIRED. THIS PROCEDURE WILL RESULT IN THE ERROR EXITING TO THE PRIMARY TEST REGARDLESS OF WHERE THE ERROR IS DETECTED. IF DESIRED, SR BIT 10 MAY BE SET WHICH RESULTS IN THE ERROR EXITING TO "PC"+1. IN THIS CASE THE SPECIFIED "LOOP N" DOES NOT NECESSARILY APPLY.

ERROR	PC	DR	TST	FAILURE	RECOVERY
1A	1035	--	1	PLCF DOES NOT CLEAR FLAG OR PLSF SKIPS WHEN FLAG CLEAR.	LOOP1 (CYCLE IN TEST1 ON PLPU-PLCF CHECK)
	1053				
	1075				
	1120		2		
	1136				
	1160				
	1212		3		
	1464		6		
	1505				
	1542				

TABLE 1 (CONT'D)

ERROR	PC	DR	TST	FAILURE	RECOVERY
4A	1274	--	4	CFLR IOT WITH AC=31 SETS FLAG PRIOR TO 50MS.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
4B	1300	--	4	CFLR IOT WITH AC=31 DOES NOT SET FLAG IN 95MS.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
4C	1317	--	4	CFLR IOT WITH AC=32 SETS FLAG PRIOR TO 50MS.	LOOP2 (CYCLE IN TEST 4 ON CFLR32 CHECK.)
4D	1323	--	4	CFLR IOT WITH AC=32 DOES NOT SET FLAG IN 95MS.	LOOP2 (CYCLE IN TEST 4 ON CFLR32 CHECK.)
4E	1266	--	4	CFLR IOT DOES NOT INITIALLY CLEAR FLAG OR PLSF SKIPS WHEN FLAG IS CLEAR.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
	1311				
	1413		5		
5A	1420	NN	5	CFLR IOT WITH AC=NN SETS FLAG PRIOR TO 5MS.	LOOP1 (CYCLE IN TEST 5 ON PRESENT CFLR NN CHECK.)
5B	1424	NN	5	CFLR IOT WITH AC=NN DOES NOT SET FLAG IN 10MS.	LOOP1 (CYCLE IN TEST 5 ON PRESENT CFLR NN CHECK.)
6A	1501	--	6.	INTERRUPT DOES NOT OCCUR AFTER PLSE IOT AND FLAG SET.	LOOP1 (CYCLE IN TEST 6 ON PLSE CHECK.)

KEY TO TABLE 2

DIRECTION CONSTANTS

COORDINATE
TERMINOLOGY:+Y CORRESPONDS TO PEN LEFT.
+X CORRESPONDS TO DRUM DOWN.DIRECTION
TERMINOLOGY:N CORRESPONDS TO PEN LEFT.
E CORRESPONDS TO DRUM DOWN.
S CORRESPONDS TO PEN RIGHT.
W CORRESPONDS TO DRUM UP.
L CORRESPONDS TO HALF VECTORS
FOR 24 VECTOR PLOTTERS.NOTE THAT DIRECTION CONSTANTS 5X AND 7X APPLY TO 24 VECTOR
PLOTTERS ONLY.ALL BLANKS ARE INTENTIONAL AND IMPLY THAT THE PARTICULAR
FUNCTION IS NON-EXISTENT FOR THAT TYPE PLOTTER.THE 500 SERIES PLOTTERS USE THE PLPU(6503) AND PLPD(6505) IOT'S
FOR PEN COMMANDS VICE DIRECTION CONSTANTS.

DIRECTION CONSTANTS TABLE 2

**NOTE: 500 SERIES FORMAT ALSO APPLIES TO HOUSTON DP10/EDP10

COORDINATE TERMINOLOGY	DIRECTION TERMINOLOGY	500 SERIES	600 SERIES	700 SERIES	REMARKS
+Y	N	20	10	10	
+Y +X	NE	30	11	11	
+X	E	10	12	12	
-Y +X	SE	50	13	13	
-Y	S	40	14	14	
-Y -X	SW	44	15	15	
-X	W	04	16	16	
+Y -X	NW	24	17	17	
+Y +X/2	NNE		72	72	---
+Y/2 +X	ENE		70	70	2
-Y/2 +X	ESE		74	74	4
-Y +X/2	SSE		76	76	
-Y -X/2	SSW		77	77	V
-Y/2 -X	WSW		75	75	E
+Y/2 -X	WNW		71	71	C
+Y -X/2	NNW		73	73	T
+Y/2	LN		50	50	O
+Y/2 +X/2	LNE		51	51	R
+X/2	LE		52	52	
-Y/2 +X/2	LSE		53	53	O
-Y/2	LS		54	54	N
-Y/2 -X/2	LSW		55	55	L
-X/2	LW		56	56	Y
+Y/2 -X/2	LNW		57	57	---

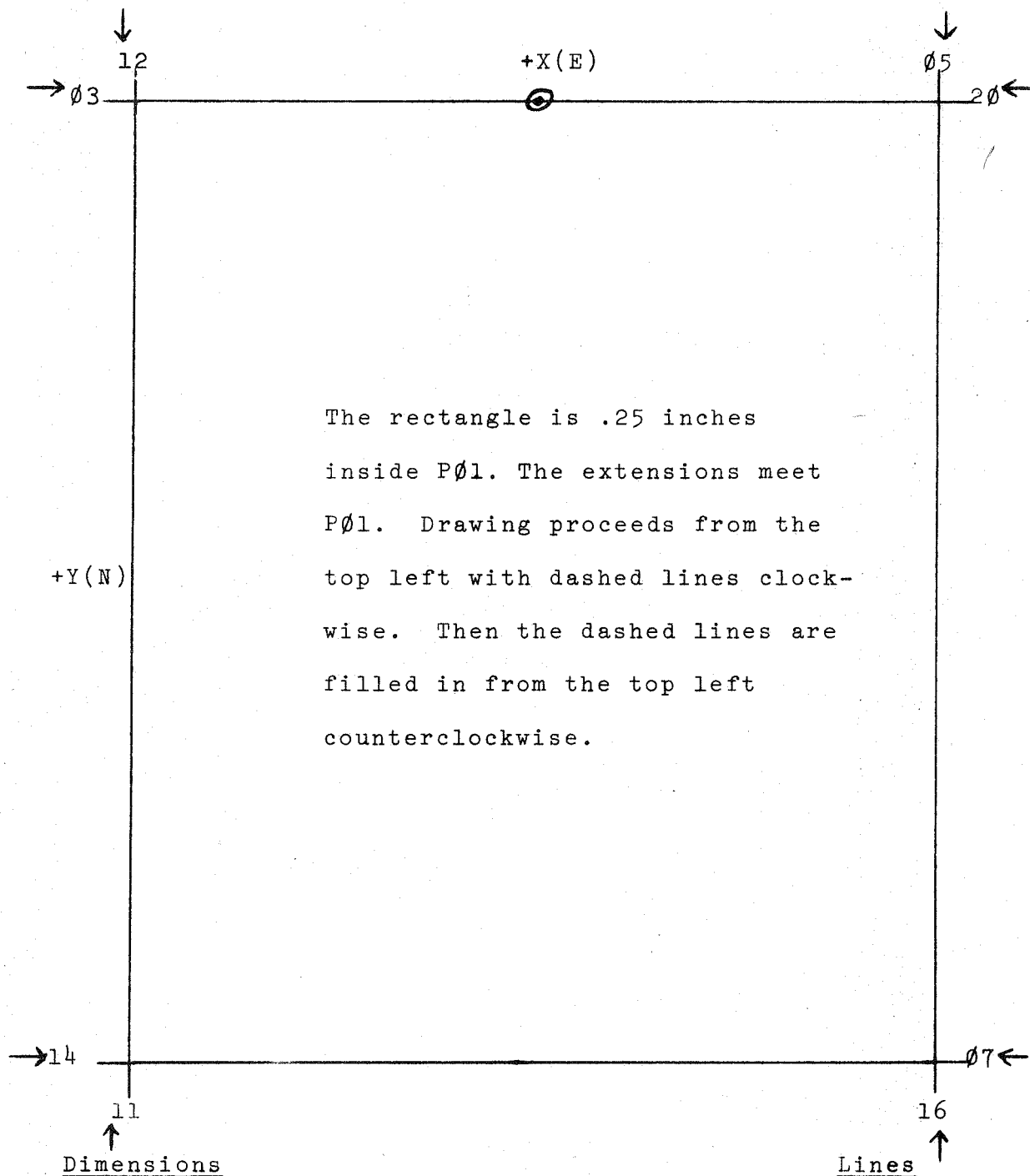
K TO FIGURE 1.

FIGURE 1 CONTAINS A REASONABLE FACSIMILE OF PATTERNS 01 THROUGH 06 (P01--P06). ONLY THE NECESSARY DIMENSIONS ARE LISTED.

ALL PATTERN NUMBERS AND LINE NUMBERS CONSIST OF TWO OCTAL DIGITS. LINES WHICH ARE TRACED WITH THE PEN UP ARE LISTED UNDER EACH PATTERN ALONG WITH THEIR PARTICULAR FUNCTION. ARROWS INDICATE THE DIRECTION IN WHICH THE LINES ARE DRAWN.

IF IT IS DESIRED TO UTILIZE P07, NOT SHOWN IN THIS FIGURE, CONSULT THE P07 LIST AT THE END OF THE PROGRAM LISTING FOR LINE FUNCTIONS.

.25 Inch Dashed Lines Filled In



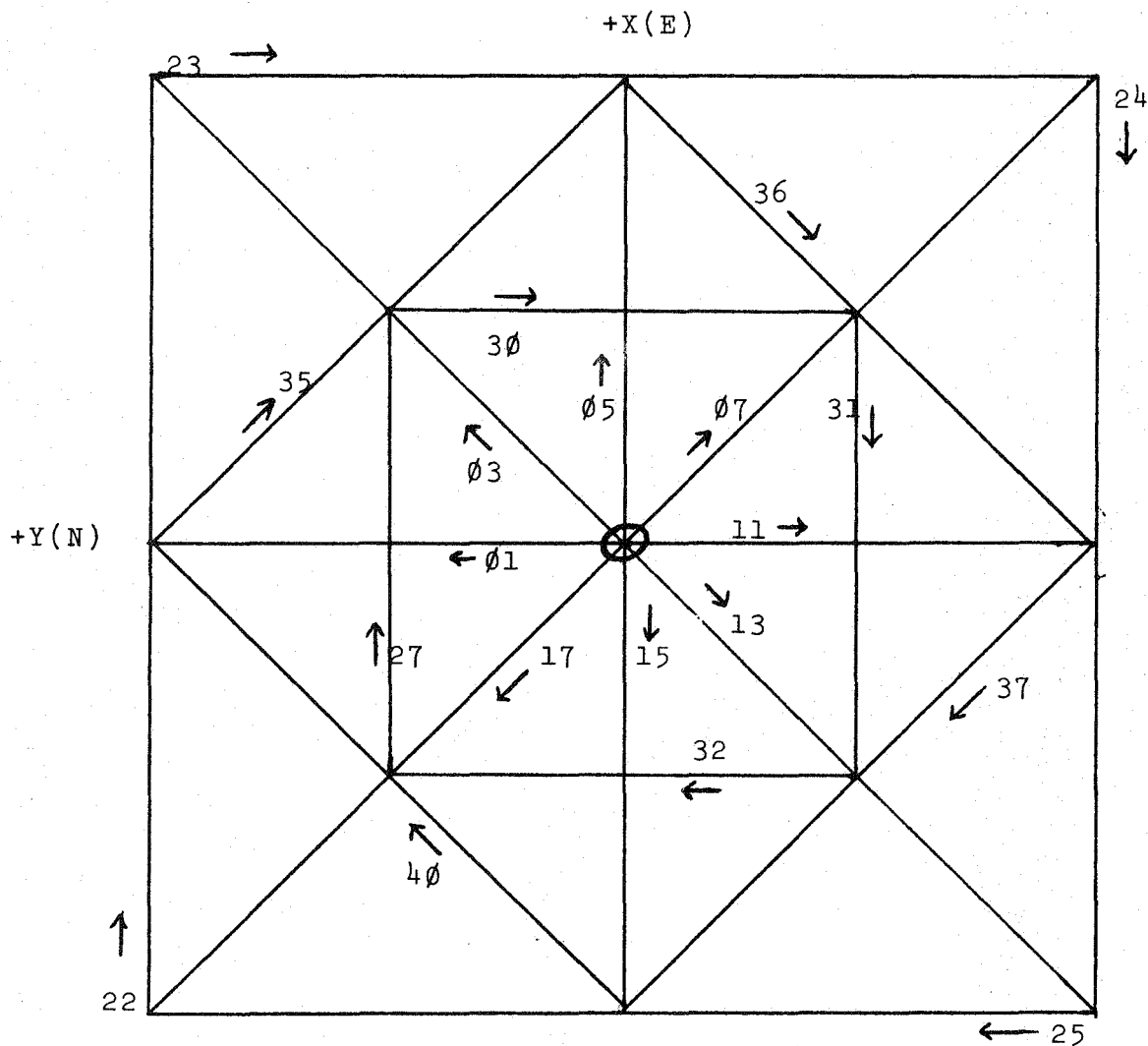
Including extensions the dimensions are the same as those of P01.

01-02 orient to start of 03.
04,06,10,13, 15, 17, 21all orient to beginning of next line
22 orients to P02 origin.
23 orients to P03 origin.

Figure 1-2

P04 (8 Vector only)

All Vector with Concentric and
Oblique Squares



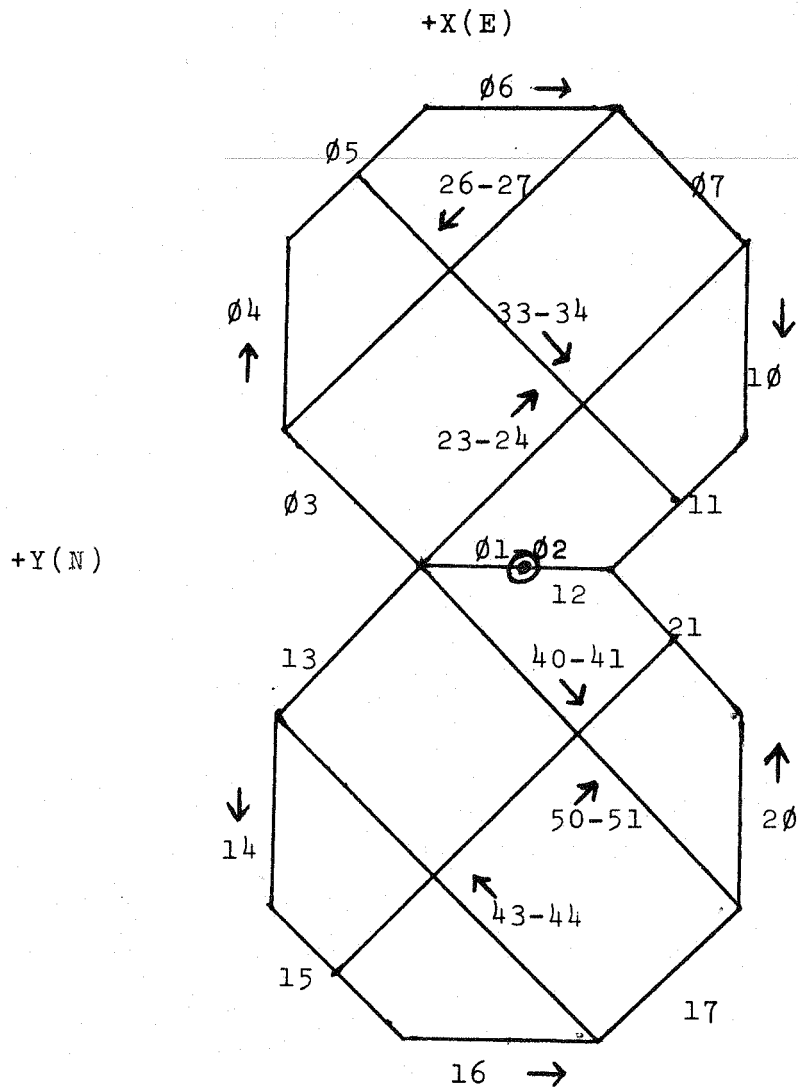
Dimensions

The outer square is
8 inches and is not
dependent on plotter size.

Lines

02, 04, 06, 10, 12, 14, 16 and 20 are
returns to origin after each
preceding line.
21 orients to start of 22.
26 orients to start of 27.
33-34 orients to start of 35.
41 orients to P04 origin
42 orients to P05 origin.

Figure 1-4-8



Dimensions

Diagonals are 6.05 inches.

Each octagon segment is
2 inches.

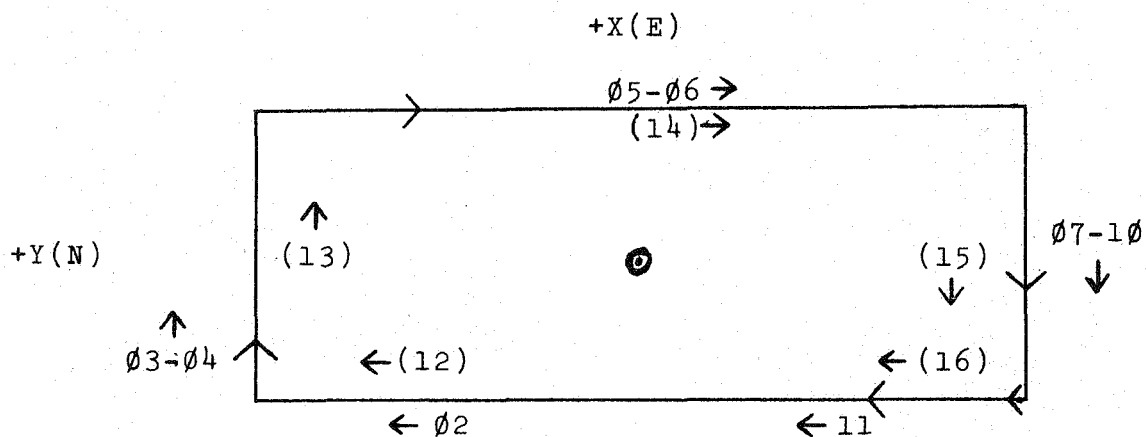
All dimensions are independent
of plotter size

Lines

22 orients to 23.
25 orients to 26.
30-32 orient to 33.
35-37 orient to 40.
42 orients to 43.
45-47 orient to 50.
52-55 orient to P05 origin.
56 orients to P06 origin

Figure 1-5-8

Rapid Arrows on a Rectangle.



Dimensions

Rectangle = 6 x 2 inches.

Arrows are 0.1 inches apart and each arrow diagonal is 0.1 inches.

Dimensions are independent of plotter size.

Lines

Lines in parentheses refer to special LID's which generate arrows.

01 orients to 02.

17 orients to P06 origin.

20 orients to P03 origin.

Figure 1-6

/DR: CONTENTS OF AC (OR DR) FOR PLLR OR CFLR FAILURES.

/UNIQUE ERROR ROUTINES
/GENERAL ERROR SERVICE ROUTINES

0017 0000 AUTO7, 0 /MSGN USES.

0020 #20 /POINTERS

/XY8E IOT SUBROUTINE POINTERS.

0020 4400 SPLCEP, SPLCE
 0021 4404 SPLSFP, SPLSF
 0022 4411 SPLCFP, SPLCF
 0023 4415 SPLPUP, SPLPU
 0024 4421 SPLLRP, SPLLR
 0025 4425 SPLPDP, SPLPD
 0026 4431 SCFLRP, SCFLR
 0027 4435 SPLSEP, SPLSE
 0030 4165 CRLFP, CRLF
 0031 4232 MSGNP, MSGN
 0032 4224 BELLP, BELL
 0033 4217 READNP, READN
 0034 4140 DSEP, DSE
 0035 4156 TYPEP, TYPE
 0036 4274 DELNMP, DELNMS

0037 1600 SETIPP, SETIP
 0040 1633 ERR1AP, ERR1A
 0041 2102 ERR4EP, ERR4E
 0042 2237 ERRORP, ERRORS
 0043 2353 QCONTP, QCONT

/GENERAL PEN (BEAM) SUBROUTINE POINTERS. DEFINED BY SERIES PLOTTER
 /CONNECTED.

0044 0044 PENS, .
 0045 0000 CPUP, 0 /PEN UP OR BEAM OFF. (NO FLAG SENSE).
 0046 0000 CPDP, 0 /PEN DOWN OR BEAM ON. (NO FLAG SENSE.)
 0047 0000 DPUP, 0 /PEN UP OR BEAM OFF. (FLAG SENSE.)
 0050 0000 DPDP, 0 /PEN DOWN OR BEAM ON. (FLAG SENSE.)

0051 0000 SAVEAC, 0
 0052 0000 SAVED, 0
 0053 0000 ALLSET, 0 /SET WHEN INIT DONE.
 0054 0000 LOPFLG, 0 /SET FOR CL+ OR CCL+.
 0055 0000 PENFLG, 0 /SET WHEN PEN DOWN.
 0056 0000 FLAG24, 0 /SET FOR 24 VECTOR.
 0057 0000 FLGZIP, 0 /SET FOR 700 SERIES.
 0060 0000 ACTFLG, 0 /SET WHEN DISPLAY COMMANDS BEING EXECUTED
 0061 0000 MSGFLG, 0 /SET WHEN ERROR HEADING ALREADY DISPLAYED.
 0062 0000 INTFLG, 0 /SET IF INTERRUPT OCCURRED DURING CONTROL TESTS.
 0063 0000 BUFF1, 0 /USED FOR ERROR CODE FILL IN.
 0064 0000 BUFF2, 0 /USED FOR PC FILL IN.
 0065 5555 BUFF3, 5555 /USED FOR DR FILL IN. SET TO DASHES IF N/A.

0066 0006 K6, 6

0153	4426	SPLPD+1
0154	4432	SCFLR+1
0155	4436	SPLSE+1
0156	0770	K770, 770
0157	7007	K7007, 7007
0160	0000	DEVCOT, 0
0161	7550	K818L, -230
0162	4520	M304P, M304

0200 *200

/JUMP TABLE.

0200	5230	JUMTAB, JMP	EXEC	/RUN COMPLETE DIAGNOSTIC.
0201	5236	JMP	EXEC2	/DITTO WITHOUT TST0 AND INIT.
0202	4777	JMS	INIT	/INIT AND HALT.
0203	7402	HLT		
0204	5776	JMP	DISPLA	/DISPLAY TEST MONITOR.
0205	4775	JMS	TST0	/TEST 0 AND HALT.
0206	7402	HLT		
0207	5205	JMP	.-2	/REPEAT TEST 0.
0210	4774	JT1, JMS	TST1	/TEST 1 AND HALT
0211	7402	HLT		
0212	4773	JMS	TST2	/TEST 2 AND HALT.
0213	7402	HLT		
0214	4772	JMS	TST3	/TEST 3 AND HALT.
0215	7402	HLT		
0216	4771	JMS	TST4	/TEST 4 AND HALT
0217	7402	HLT		
0220	4770	JMS	TST5	/TEST 5 AND HALT
0221	7402	HLT		
0222	4767	JMS	TST6	/TEST 6 AND HALT
0223	7402	HLT		
0224	5210	JMP	JT1	/GO TO TEST 1 AND HALT.
0225	4122	JMS	DEVCOD	/CHANGE DEVICE CODES.
0226	7402	HLT		
0227	5226	JMP	.-1	

/XY8E DIAGNOSTIC PROGRAM EXECUTIVE.

/1. RUNS COMPLETE CONTROL TEST (MAKES 10 COMPLETE PASSES.)

/THE CONTROL TEST CONSISTS OF THE FOLLOWING (1 PASS)

/TEST 0 ONLY ONCE (RUN ON FIRST PASS ONLY)

/TESTS 1 THRU 6 10 TIMES EACH.

/2. RUNS DISPLAY TEST UNTIL "LT=" COMMAND AT WHICH POINT

/STEP 1 IS REPEATED WITHOUT TEST 0, AND SO ON.

/3. THE TTY BELL SOUNDS ON EACH PASS OF THE CONTROL TEST.

/4. SR11 SET LOOPS ON WHOLE CONTROL TEST (WITHOUT TEST 0).

/5. "CONTROL TEST COMPLETE" DISPLAYED AT END OF STEP 1.

0230	4766	EXEC, JMS	PATCH	
0231	4775	JMS	TST0	/RUN TEST 0 ONCE.
0232	4430	JMS I	CRLF	
0233	1520	TAD	MSG1P	/PRINT TITLE MSG.

0320	4600
0321	4761
0322	0000
0323	0000
0324	5002

MSG1P, MSG1
MSG14P, MSG14
EXECNT, Ø
TSTTAL, Ø
MSG17P, MSG17

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/*****
/LOCATIONS 330 THRU 362 USED FOR ASSEMBLY STORAGE OF PROGRAM DIRECTORY.
/*****/

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/PROGRAM INITIALIZATION ROUTINE.

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/1.  CARRIES ON A DIALOGUE WITH THE USER TO DETERMINE THE SPECIFICATIONS
/OF THE PLOTTER TO BE TESTED.  THE USER REPLIES WITH NUMERIC CODES AS
/DEFINED BELOW.

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/2. WHEN ALL INPUTS HAVE BEEN RECEIVED THE INFORMATION IS ECHOED ON
/THE TTY IN DECODED FORM FOR USER VERIFICATION.  USER TYPES "Y" OR "N"
/(FOR YES OR NO) IN ANSWER TO "CORRECT?" IF RESPONSE IS "N" THE DIALOGUE
/IS REPEATED.

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13. ALL LISTS ARE MODIFIED TO ENSURE PROPER FORMATTING DEPENDENT
ON PLOTTER SPECIFICATIONS.

/IF AT ANY TIME THE INCREMENT SIZE IS MANUALLY CHANGED, IT WILL BE
/NECESSARY TO GO THROUGH THE COMPLETE DIALOGUE AGAIN.

THE FOLLOWING NUMERIC CODES ARE TO BE UTILIZED BY USER IN RESPONSE
TO THE DIALOGUE.

/ PLOTTER SERIES CODES.

/"1" 300 SERIES

1"2" 600 SERIES

/"3" 700 SERIES

1/PLOTTER TYPE CODES.

/"1" 11 INCH DRUM.

28.55 INCH DRUM (TREATED AS 28 INCHES WIDE)

31 X 54 INCH FLATBED

1"4" 54 X 72 INCH FLATBED

/VECTOR TYPE CODES:

/"1" 8 VECTOR

/"2" 24 VECTOR

0443	1412	TAD I	AUTO2	
0444	3361	DCA	INITEM	
0445	1761	TAD I	INITEM	
0446	4431	JMS I	MSGNP	
0447	2332	ISE	INITAL	
0450	5237	JMP	INIL2	
0451	4430	JMS I	CRLFP	
0452	4430	JMS I	CRLFP	
0453	1334	TAD	MSG13P	
0454	4431	JMS I	MSGNP	/REQUEST USER DECISION AS
0455	4432	JMS I	BELLP	/TO CORRECTNESS OF DATA.
0456	4433	JMS I	READNP	/LOOK FOR Y=YES; N=NO
0457	0073	AND	K1	
0460	7650	SNA CLA		/INFO CORRECT?
0461	5201	JMP	INIT+1	/NO. REQUEST INFO AGAIN.
0462	4430	JMS I	CRLFP	
0463	4434	SERIES, JMS I	DSEP	/IS SERIES 500?
0464	0550	PS		
0465	5273	JMP	.,+6	
0466	1777'	TAD	P500	/YES. SET UP 500 SERIES
0467	3010	DCA	AUTO0	/SUBROUTINE POINTERS FOR
0470	1776'	TAD	DK500	/TRANSFER TO STACK.
0471	3012	DCA	AUTO2	
0472	5311	JMP	TRSER	/GO TO TRANSFER OF POINTERS.
0473	1775'	TAD	P678	/SERIES IS 600 OR 700.
0474	3010	DCA	AUTO0	/SET UP 600-700 SERIES SUBROUTINE
0475	1774'	TAD	DK678	/POINTERS FOR TRANSFER TO
0476	3012	DCA	AUTO2	/STACK.
0477	4434	JMS I	DSEP	/IS SERIES 600?
0500	0550	PS		
0501	7610	SKP CLA		/NO.
0502	7000	NOP		/YES.
0503	7000	NOP		
0504	4434	JMS I	DSEP	/IS SERIES 700?
0505	0550	PS		
0506	7610	SKP CLA		/NO. CLEAR ZIP FLAG.
0507	7240	CLA CMA		/YES. SET ZIP FLAG.
0510	3057	DCA	FLGZIP	
0511	1044	TRSER, TAD	PENS	/TRANSFER PEN SUBROUTINE POINTERS.
0512	3011	DCA	AUTO1	
0513	1105	TAD	M4	
0514	3332	DCA	INITAL	
0515	1410	TAD I	AUTO0	
0516	3411	DCA I	AUTO1	
0517	2332	ISE	INITAL	
0520	5315	JMP	.-3	
0521	1773'	TAD	DKAYS	/TRANSFER VARIABLE DIRECTION
0522	3013	DCA	AUTO3	/CONSTANTS.
0523	1106	TAD	M10	
0524	3332	DCA	INITAL	
0525	1412	TAD I	AUTO2	
0526	3413	DCA I	AUTO3	
0527	2332	ISE	INITAL	
0530	5025	JMP	.-3	

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0616 3346          DCA      CONCNT
0617 1410          TAD I    AUTO0
0620 3347          DCA      CONTEM
0621 1347          CONL2,  TAD      CONTEM
0622 3411          DCA I    AUTO1
0623 1347          TAD      CONTEM
0624 7104          CLL RAL
0625 3347          DCA      CONTEM
0626 2346          ISZ      CONCNT
0627 5221          JMP      CONL2
0630 2345          ISZ      CONL1
0631 5215          JMP      CONL1
0632 4434          VECTOR, JMS I  DSZP
0633 0552          VT
0634 7610          SKP CLA
0635 7610          SKP CLA
0636 7240          CLA CMA
0637 3056          DCA      FLAG24
0640 4434          INCR,  JMS I  DSZP
0641 0553          IT
0642 5256          JMP      INCR1
0643 1775'         TAD      METER
0644 3010          DCA      AUTO0
0645 1344          TAD      LGTP
0646 3011          DCA      AUTO1
0647 1350          TAD      M20
0650 3345          DCA      CONLAL
0651 1410          TAD I    AUTO0
0652 3411          DCA I    AUTO1
0653 2345          ISZ      CONLAL
0654 5251          JMP      .-3
0655 5307          JMP      PATTR
0656 1774'         INCR1,  TAD      INSTP
0657 3010          DCA      AUTO0
0660 4434          JMS I    DSZP
0661 0553          IT
0662 5270          JMP      .+6
0663 1410          TAD I    AUTO0
0664 3302          DCA      INCR11
0665 1410          TAD I    AUTO0
0666 3303          DCA      INCR12
0667 5273          JMP      .+4
0670 2010          ISZ      AUTO0

```

/SET 24 VECTOR FLAG IF
/PLOTTER IS 24 VECTOR TYPE.

/SET UP LENGTH CONSTANTS
/ACCORDING TO INCREMENT SIZE
/USING .0025 INCHES AS BASIC UNLESS
/INCREMENT SIZE IS 0.1 MM.
/TRANSFER METRIC CONSTANTS.

/INCREMENT SIZE IS TYPE 2,3,4, OR 5.
/TRANSFER SCALED CONSTANTS

```

0671 2010          ISZ      AUTO0
0672 5260          JMP      INCR1+2
0673 1773'         TAD      INCH
0674 3010          DCA      AUTO0
0675 1344          TAD      LGTP
0676 3011          DCA      AUTO1
0677 1352          TAD      M50
0700 3345          DCA      CONLAL
0701 1410          TAD I    AUTO0
0702 0000          INCR11, 0

```

/SCALE FACTOR INSTRUCTIONS

0777 5257
1000 *1000

/START CONTROL TESTS.

/TEST0: TEST OF THE ABILITY OF THE SIGNAL "INITIALIZE" TO CLEAR THE
/PLOTTER FLAG AND SET INTERRUPT ENABLE.
/THIS TEST IS RUN ONLY ONCE PRIOR TO ANY TYPEOUTS. IF AN
/ERROR OCCURS THE PROGRAM WILL HALT. IF THE FLAG WILL NOT
/SET OR PLSF WILL NOT SKIP, THE PROGRAM WILL HANG UP.
/ONLY AFTER THIS TEST PASSES WILL THE TITLE MESSAGE BE
/PRINTED.
/ALL ERROR RECOVERIES SHOULD BE MADE BY DEPRESSING START
/TO GENERATE ANOTHER "INITIALIZE" SIGNAL.
/THIS TEST MAY BE INHIBITED BY SETTING SR0.

1000	0000	TST0.	0	
1001	7200		CLA	/POP8-E USERS MAY INSERT CAF HERE.
1002	4437		JMS I SETIPP	/SET UP INTERRUPT LINK.
1003	7604		LAS	
1004	0071		AND K4	
1005	7640		SEA CLA	
1006	5600		JMP I TST0	/TEST 0 INHIBITED.
1007	4421		PLSF	
1010	5213		JMP .+3	
1011	7402	ERR0A.	HLT	/INITIALIZE DOES NOT CLEAR
1012	5201		JMP TST0+1	/FLAG OR PLSF SKIPS WHEN
				/FLAG IS CLEAR.
1013	4424		PLLR	/TRY TO SET FLAG.
1014	4421	HANGUP.	PLSF	/IF PROGRAM HANGS UP PLLR
1015	5214		JMP .-1	/((DR=0) DOES NOT SET FLAG.
1016	7200		CLA	/FLAG SETS. CHECK INTERRUPT.
1017	3062		DCA INTFLG	
1020	6001		ION	
1021	7000		NOP	
1022	6002		IOF	
1023	1062		TAD INTFLG	
1024	7640		SEA CLA	
1025	5600		JMP I TST0	/TST0 PASSES. EXIT.
1026	7402	ERR0B.	HLT	/INITIALIZE DOES NOT SET INT.
1027	5201		JMP TST0+1	/ENABLE OR (FLAG AND INT ENABLE)
				/DO NOT YIELD INT. REQUEST.

/TEST1: FLAG TEST FOR PLPU AND PLPD IOT'S.
/CHECKS THE FOLLOWING:
/1. PLCF CLEARS FLAG.
/2. PLSF SKIPS ONLY WHEN FLAG SET.
/3. PLPU AND PLPD DO NOT SET FLAG PRIOR TO 70 MS.
/4. PLPU AND PLPD SET FLAG AT 70 MS.
/ERRORS 1A THRU 1E ARE DETECTED.
/LOOP 1 CYCLES ON PLPU - PLCF CHECK.
/LOOP 2 CYCLES ON PLPD - PLCF CHECK.
/LOOP 3 CYCLES ON WHOLE TEST.

```

1104 7640      SZA CLA
1105 5231      JMP      T1L1L3      /YES. LOOP 3.

```

```

/
/END LOOP 3.
/

```

```

1106 5630      JMP I   TST1      /EXIT.

```

```

1107 1646      ERR1BP, ERR1B
1110 1661      ERR1CP, ERR1C
1111 1674      ERR1DP, ERR1D
1112 1707      ERR1EP, ERR1E

```

```

/TEST2: FLAG TEST FOR PLLR WITH DIRECTION REGISTER = 31 AND
/32. THIS IS EQUIVALENT FOR 600 - 700 SERIES TO PLPU AND
/PLPD.
/THE FOLLOWING CHECKS ARE MADE.
/1. PLCF CLEARS FLAG.
/2. PLSF SKIPS ONLY WHEN FLAG SET.
/3. PLLR 31 AND 32 DO NOT SET FLAG PRIOR TO 70 MS.
/4. PLLR 31 AND 32 SET FLAG AT 70 MS.
/ERRORS 1A, AND 2A THRU 2D ARE DETECTED.
/LOOP 1 CYCLES ON PLLR31 - PLCF CHECK.
/LOOP 2 CYCLES ON PLLR32 - PLCF CHECK.
/LOOP 3 CYCLES ON WHOLE TEST.

```

```

1113 0000      TST2, 0
/
/START LOOP 1 AND 3.
/

```

```

1114 7300      T2L1L3, CLA CLL      /START LOOP1 AND 3.
1115 4422      PLCF                /CLEAR FLAG.
1116 4421      PLSF                /FLAG CLEAR.
1117 7410      SKP
1120 4440      JMS I   ERR1AP      /NO. FLAG SET OR ILLEGAL SKIP.
1121 4777      JMS     CPU8        /DO PLLR31.
1122 1115      TAD     K70MIN      /WAIT 60 MS.
1123 4436      JMS I   DELNMP
1124 4421      PLSF                /FLAG SET?
1125 7410      SKP
1126 4772      JMS I   ERR2AP      /YES. FLAG SET PRIOR TO 70 MS.
1127 1116      TAD     K70MAX      /WAIT 20 MORE MS.
1130 4436      JMS I   DELNMP
1131 4421      PLSF                /FLAG SET?
1132 4773      JMS I   ERR2BP      /NO. FLAG NOT SET AT 70 MS.
1133 4422      PLCF                /CLEAR FLAG.
1134 4421      PLSF                /FLAG CLEAR?
1135 7410      SKP
1136 4440      JMS I   ERR1AP      /NO. FLAG SET OR ILLEGAL SKIP.
1137 7604      LAS
1140 0070      AND     K400
1141 7640      SZA CLA
1142 5314      JMP      T2L1L3      /YES. LOOP 1.

```

```

/
/END LOOP 1.

```

```

1200 0000 TST3, 0
/
/START LOOP 2.
/
1201 7300 T3L2, CLA CLL
1202 3253 DCA KLDRN
1203 1112 TAD M31
1204 3257 DCA T3M31
1205 1113 TAD M100
1206 3254 DCA T3CNT
/
/START LOOP 1.
/
1207 4422 T3REP, PLCF
1210 4421 PLSF
1211 7410 SKP
1212 4440 JMS I ERR1AP
1213 1253 TAD KLDRN
1214 4424 PLLR
1215 7200 CLA
1216 1117 TAD K5MIN
1217 4436 JMS I DELNMP
1220 4421 PLSF
1221 7410 SKP
1222 4655 JMS I ERR3AP
1223 1120 TAD K5MAX
1224 4436 JMS I DELNMP
1225 4421 PLSF
1226 4656 JMS I ERR3BP
1227 7604 LAS
1230 0070 AND K400
1231 7640 SZA CLA
1232 5207 JMP T3REP
/
/END LOOP 1.
/
1233 2253 ISZ KLDRN
1234 2257 ISZ T3M31
1235 5244 JMP .+7
1236 2253 ISZ KLDRN
1237 2253 ISZ KLDRN
1240 2253 ISZ KLDRN
1241 2254 ISZ T3CNT
1242 2254 ISZ T3CNT
1243 2254 ISZ T3CNT
1244 2254 ISZ T3CNT
1245 5207 JMP T3REP
1246 7604 LAS
1247 0101 AND K200
1250 7640 SZA CLA
1251 5201 JMP T3L2
/
/END LOOP 2.
/

/SET N = 0
/SET COUNTER TO SKIP DR = 31, 32 OR 33.
/SET MAIN COUNT = -64

/CLEAR FLAG.
/FLAG CLEAR?

/NO. FLAG SET OR ILLEGAL SKIP
/PUT CURRENT N IN AC.
/LOAD DIRECTION REGISTER.

/WAIT 4 MS.

/FLAG SET?

/YES. FLAG SET PRIOR TO 5 MS.
/WAIT 2 MORE MS.

/FLAG SET?
/NO. FLAG NOT SET AT 5 MS.
/LOOP 1?

/YES. LOOP 1.

/UPDATE N.
/N = 31?

/YES. BYPASS N=31, 32 AND 33.

/UPDATE MAIN COUNT
/NOT DONE. CONTINUE.
/LOOP 2?

/YES. LOOP2.

```

```

1312 7200      CLA
1313 1115      TAD      K70MIN      /WAIT 60 MS.
1314 4436      JMS I    DELNMP

1315 4421      PLSF      /FLAG SET?
1316 7410      SKP
1317 4737      JMS I    ERR4CP      /YES. FLAG SET PRIOR TO 70 MS.
1320 1116      TAD      K70MAX      /WAIT 20 MORE MS.
1321 4436      JMS I    DELNMP
1322 4421      PLSF      /FLAG SET?
1323 4740      JMS I    ERR4CP      /NO. FLAG NOT SET AT 70 MS.
1324 7604      LAS
1325 0101      AND      K200
1326 7640      SZA CLA
1327 5305      JMP      T4L2      /YES. LOOP 2.

```

/END LOOP 2.

```

1330 7604      LAS      /LOOP 3?
1331 0067      AND      K100
1332 7640      SZA CLA
1333 5261      JMP      T4L1L3      /YES. LOOP 3.

```

/END LOOP 3.

```

1334 5660      JMP I    TST4      /EXIT.
1335 2026      ERR4AP, ERR4A
1336 2041      ERR4BP, ERR4B
1337 2054      ERR4CP, ERR4C
1340 2067      ERR4DP, ERR4D

```

1400 *1400

```

/TEST5: FLAG TEST FOR CFLR WITH DIRECTION REGISTER = N; WHERE
/N = 0 THRU 77 EXCEPT FOR 31,32 AND 33. THE METHOD IS SIMILAR TO TEST 3.
/THE FOLLOWING CHECKS ARE MADE:
/1. CFLR INITIALLY CLEARS FLAG.
/2. PLSF SKIPS ONLY WHEN FLAG SET.
/3. CFLR N DOES NOT SET FLAG PRIOR TO 5 MS.
/4. CFLR N SETS FLAG AT 5 MS.
/ERRORS 4E, AND 5A THRU 5B ARE DETECTED.
/LOOP 1 CYCLES ON PRESENT CFLR N.
/LOOP 2 CYCLES ON WHOLE TEST.

```

1400 0000 TST5, 0

/START LOOP 2.

```

1401 7300      T5L2.    CLA CLL
1402 3251      DCA      KCFLRN      /SET N = 0
1403 1112      TAD      M31          /SET COUNTER TO SKIP DR = 31, 32, 33.
1404 3252      OCA      T5M31
1405 1113      TAD      M100        /SET MAIN COUNT = -64

```

/4. PLCE CLEARS INTERRUPT ENABLE.
 /5. INTERRUPT OCCURS ONLY AFTER PLSE AND FLAG SET.
 /ERRORS 1A, AND 6A THRU 6E ARE DETECTED.
 /LOOP 1 CYCLES ON PLSE CHECK.
 /LOOP 2 CYCLES ON PLCE CHECKS.
 /LOOP 3 CYCLES ON WHOLE TEST.

1456	0000	TST6, 0		
1457	4437	JMS I SETIPP	/SET INTERRUPT LINK.	
/				
/START LOOP 1 AND 3.				
/				
1460	4420	T6L1L3, PLCE	/DISABLE INT	
1461	4422	PLCF	/CLEAR FLAG.	
1462	4421	PLSF	/FLAG CLEAR?	
1463	7410	SKP		
1464	4440	JMS I ERR1AP	/NO. FLAG NOT CLEAR OR ILLEGAL SKIP.	
1465	4426	CFLR	/LOAD DR.	
1466	1121	TAD K5ALL	/WAIT 10 MS.	
1467	4436	JMS I DELNMP		
1470	4421	PLSF	/FLAG SET?	
1471	4763	JMS I ERR6EP	/NO. FLAG NOT SET AT 5 MS.	
1472	3062	DCA INTFLG	/SET FLAG = 0	
1473	4427	PLSE	/ENABLE INT	
1474	6001	ION	/INT. ON	
1475	7000	NOP		
1476	6002	IOF	/KILL INT	
1477	1062	TAD INTFLG	/DID INT. OCCUR	
1500	7700	SMA CLA		
1501	4764	JMS I ERR6AP	/NO INT AFTER PLSE AND FLAG SET	
1502	4422	PLCF	/CLEAR FLAG.	
1503	4421	PLSF	/FLAG CLEAR?	
1504	7610	SKP CLA		
1505	4440	JMS I ERR1AP	/NO. FLAG NOT CLEAR OR ILLEGAL SKIP	
1506	3062	DCA INTFLG	/SET FLAG TO 0	
1507	6001	ION	/SET INT.	
1510	7000	NOP		
1511	6002	IOF	/KILL INT	
1512	1062	TAD INTFLG	/INT. OCCUR?	
1513	7640	SEA CLA		
1514	4765	JMS I ERR6BP	/YES. INT AFTER PLSE AND FLAG CLEAR.	
1515	7604	LAS	/LOOP 1?	
1516	0070	AND K400		
1517	7640	SEA CLA		
1520	5260	JMP T6L1L3		
/				
/END LOOP 1.				
/				
/START LOOP 2.				
/				
1521	4427	T6L2, PLSE	/ENABLE INT	
1522	4426	CFLR	/LOAD DR.	

```

1604 1207      TAD      CTINTP
1605 3002      DCA      2
1606 5600      JMP I    SETIP

```

/INTERRUPT SERVICE FOR CONTROL TESTS.

```

1607 1610      CTINTP, .+1
1610 3051      DCA      SAVEAC
1611 7010      RAR
1612 3052      DCA      SAVEL
1613 6031      KSF
1614 5217      JMP      .+3
1615 6032      KCC
1616 5226      JMP      FRT
1617 6041      TSF
1620 5223      JMP      .+3
1621 6042      TCF
1622 5226      JMP      FRT
1623 7240      CLA CMA
1624 3062      DCA      INTFLG      /SET INTFLG IF TTY NOT CAUSE
1625 5400      JMP I    0
1626 1052      FRT,    TAD      SAVEL
1627 7004      RAL
1630 1051      TAD      SAVEAC
1631 6001      ION
1632 5400      JMP I    0

```

/TAPE 3.

/ERROR ROUTINES.

/

/

/ERROR1A: PLCF DOES NOT CLEAR FLAG OR PLSF SKIPS WHEN FLAG CLEAR.

```

1633 0000      ERR1A, 0
1634 7200      CLA
1635 1245      TAD      K6101
1636 3063      DCA      BUFF1
1637 1233      TAD      ERR1A
1640 4442      JMS I    ERRORP
1641 7402      HLT
1642 4443      JMS I    QCONT
1643 5633      JMP I    ERR1A
1644 5777      JMP      TST1+1
1645 6101      K6101, 6101

```

/ERROR1B: PLPU IOT SETS FLAG PRIOR TO 70 MS.

```

1646 0000      ERR1B, 0
1647 7200      CLA
1650 1260      TAD      K6102
1651 3063      DCA      BUFF1
1652 1246      TAD      ERR1B
1653 4442      JMS I    ERRORP
1654 7402      HLT

```

```
1727 4442      JMS I  ERRORP
1730 7402      HLT
1731 4443      JMS I  QCONTP
1732 5722      JMP I  ERR2A
1733 5776      JMP    TST2+1
1734 6201      K6201, 6201
```

/ERROR2B: PLLR IOT WITH AC=31 DOES NOT SET FLAG IN 70 MS.

```
1735 0000      ERR2B, 0
1736 7200      CLA
1737 1347      TAD      K6202
1740 3063      DCA      BUFF1
1741 1335      TAD      ERR2B
1742 4442      JMS I  ERRORP
1743 7402      HLT
1744 4443      JMS I  QCONTP
1745 5735      JMP I  ERR2B
1746 5776      JMP    TST2+1
1747 6202      K6202, 6202
```

/ERROR2C: PLLR IOT WITH AC=32 SETS FLAG PRIOR TO 70 MS.

```
1750 0000      ERR2C, 0
1751 7200      CLA
1752 1362      TAD      K6203
1753 3063      DCA      BUFF1
1754 1350      TAD      ERR2C
1755 4442      JMS I  ERRORP
1756 7402      HLT
1757 4443      JMS I  QCONTP
1760 5750      JMP I  ERR2C
1761 5776      JMP    TST2+1
1762 6203      K6203, 6203
```

/ERROR2D: PLLR IOT WITH AC=32 DOES NOT SET FLAG IN 70 MS.

```
1763 0000      ERR2D, 0
1764 7200      CLA
1765 1375      TAD      K6204
1766 3063      DCA      BUFF1
1767 1363      TAD      ERR2D
1770 4442      JMS I  ERRORP
1771 7402      HLT
1772 4443      JMS I  QCONTP
1773 5763      JMP I  ERR2D
1774 5776      JMP    TST2+1
1775 6204      K6204, 6204
```

```
1776 1114
1777 1031
      2000
```

*2000

/ERROR3A: PLLR IOT WHERE AC NOT 31, 32, OR 33 SETS FLAG PRIOR TO 5 MS.

/ERROR4C: CFLR IOT WITH AC=32 SETS FLAG PRIOR TO 70 MS.

2054	0000	ERR4C,	0
2055	7200		CLA
2056	1266		TAD K6403
2057	3063		DCA BUFF1
2060	1254		TAD ERR4C
2061	4442		JMS I ERRORP
2062	7402		HLT
2063	4443		JMS I QCONTP
2064	5654		JMP I ERR4C
2065	5776		JMP TST4+1
2066	6403	K6403,	6403

/ERROR4D: CFLR IOT WITH AC=32 DOES NOT SET FLAG IN 70 MS.

2067	0000	ERR4D,	0
2070	7200		CLA
2071	1301		TAD K6404
2072	3063		DCA BUFF1
2073	1267		TAD ERR4D
2074	4442		JMS I ERRORP
2075	7402		HLT
2076	4443		JMS I QCONTP
2077	5667		JMP I ERR4D
2100	5776		JMP TST4+1
2101	6404	K6404,	6404

/ERROR4E: CFLR IOT DOES NOT INITIALLY CLEAR FLAG OR
/PLSF SKIPS WHEN FLAG CLEAR.

2102	0000	ERR4E,	0
2103	7200		CLA
2104	1314		TAD K6405
2105	3063		DCA BUFF1
2106	1302		TAD ERR4E
2107	4442		JMS I ERRORP
2110	7402		HLT
2111	4443		JMS I QCONTP
2112	5702		JMP I ERR4E
2113	5776		JMP TST4+1
2114	6405	K6405,	6405

/ERROR5A: CFLR IOT WITH AC NOT 31, 32 OR 33 SET FLAG PRIOR TO 5 MS.

2115	0000	ERR5A,	0
2116	7200		CLA
2117	1327		TAD K6501
2120	3063		DCA BUFF1
2121	1775		TAD KCFLRN
2122	3065		DCA BUFF3
2123	1315		TAD ERR5A
2124	4442		JMS I ERRORP

```

2200 0000 ERR6C, 0
2201 7200 CLA
2202 1212 TAD K6603
2203 3063 DCA BUFF1
2204 1200 TAD ERR6C
2205 4442 JMS I ERRORP
2206 7402 HLT
2207 4443 JMS I QCONTP
2210 5600 JMP I ERR6C
2211 5777 JMP TST6+1
2212 6603 K6603, 6603

```

/ERROR6D: INTERRUPT OCCURS AFTER PLCE IOT AND FLAG CLEAR.

```

2213 0000 ERR6D, 0
2214 7200 CLA
2215 1225 TAD K6604
2216 3063 DCA BUFF1
2217 1213 TAD ERR6D
2220 4442 JMS I ERRORP
2221 7402 HLT
2222 4443 JMS I QCONTP
2223 5613 JMP I ERR6D
2224 5777 JMP TST6+1
2225 6604 K6604, 6604

```

/ERROR6E: CFLR IOT WITH AC=0 DOES NOT SET FLAG IN 5 MS.

```

2226 0000 ERR6E, 0
2227 7200 CLA
2230 1236 TAD K6605
2231 3063 DCA BUFF1
2232 1226 TAD ERR6E
2233 4442 JMS I ERRORP
2234 7402 HLT
2235 5777 JMP TST6+1
2236 6605 K6605, 6605

```

/ERROR SERVICE ROUTINE. ENTERED WITH PC IN AC AND BUFFS 1 AND 3 STORED.

```

2237 0000 ERRORS, 0
2240 3064 DCA BUFF2 /STORE PC.
2241 7604 LAS /PRINTOUT INHIBITED?
2242 0262 AND K2000
2243 7640 SZA CLA
2244 7410 SKP
2245 4265 JMS ERRMSG /NO. PRINT UNIQUE ERROR MSG.
2246 1264 TAD K5555 /BLANK BUFF3.
2247 3065 DCA BUFF3
2250 7604 LAS /BELL ON ERROR?
2251 0261 AND K1000
2252 7640 SZA CLA

```

2332 5107 MSG16P, MSG16

/EDIT ROUTINE.

/ENTERED WITH 2 DIGIT OCTAL NUMBER IN AC RIGHT JUSTIFIED.
/EXITS WITH TWO NUMBERS IN AC IN PACKED ASCII.

2333	0000	EDIT,	0
2334	3347	DCA	EDTEM1
2335	1347	TAD	EDTEM1
2336	0352	AND	K70
2337	7104	CLL	RAL
2340	7006	RTL	
2341	3350	DCA	EDTEM2
2342	1347	TAD	EDTEM1
2343	0076	AND	K7
2344	1350	TAD	EDTEM2
2345	1351	TAD	K6060
2346	5733	JMP	I EDIT
2347	0000	EDTEM1,	0
2350	0000	EDTEM2,	0
2351	6060	K6060,	6060
2352	0070	K70,	70

/ROUTINE TO CHECK FOR CONTINUE EXIT. EXITS TO ENTRY +1 IF CONTINUE
/REQUESTED. OTHERWISE EXITS TO ENTRY +2.

2353	0000	QCONT,	0	
2354	7604	LAS		/CONTINUE REQUEST?
2355	0072	AND	K2	
2356	7650	SNA	CLA	
2357	2353	ISE	QCONT	/NO.
2360	5753	JMP	I QCONT	/YES.

/TAPE 4.

/DISPLAY TEST MONITOR. CONTROLLED VIA TTY WITH USER COMMANDS.
/1. MONITOR ACCEPTS TTY INPUTS AT ANY TIME DURING DISPLAY TEST.
/IF INCORRECT SYNTAX IS DETECTED, MONITOR REQUESTS NEW COMMAND STRING.
/THE FOLLOWING COMMANDS CONTROL MONITOR AND HAVE THE RESULT INDICATED.

/COMMAND (=RETURN)(=ALTMODE) RESULT

/CD-	DRAW ALL PATTERNS ONCE.
/CDIP00-	DRAW PATTERN 00 ONCE. (LIMITED TO 1 PATTERN.)
/CDIP00L00IP00L00-	DRAW PAT00 LINE00, ETC. (LIMITED TO 10 OPERANDS)
/CL-	LOOP ON ALL PATTERNS CONTINUOUSLY.
/CLIP00-	LOOP ON PATTERN 00 CONTINUOUSLY (1 PATTERN)
/CLIP00L00IP00L00-	LOOP ON PAT00 LINE00, ETC. CONTINUOUSLY (10 OPERANDS)
/C-	CONTINUE EXECUTING PREVIOUS COMMAND.
/CL-	CONTINUE LAST COMMAND BUT SET LOOP MODE.
/CD-	CONTINUE LAST COMMAND BUT CLEAR LOOP MODE.
/T-	TERMINATE DISPLAY MONITOR; GO TO CONTROL TEST EXECUTIVE.
/I-	TERMINATE DISPLAY TEST
/	MONITOR, GO TO INIT FOR
/	NEW SPECIFICATION INPUT.

2420	6002	REQUES,	IOF		/KILL INTERRUPT.
2421	4430		JMS I	CRLFP	
2422	3060		DCA	ACTFLG	/CLEAR DISPLAY ACTIVE FLAG.
2423	7200	REQU,	CLA		
2424	1232		TAD	K252	/TYPE "*" BELL.
2425	4435		JMS I	TYPEP	
2426	4432		JMS I	BELLP	
2427	5246		JMP	ALT1	/GO TO ALT MODE STRING
2430	2433	ALTP,	ALT		
2431	5016	MSG20P,	MSG20		
2432	0252	K252,	252		
		/ALT MODE STRING. CHECKS FOR ALTMODE, D,L,C,I,M, OR T.			
2433	3051	ALT,	DCA	SAVEAC	/ENTER FROM INTERRUPT.
2434	7010		RAR		/SAVE AC AND LINK.
2435	3052		DCA	SAVEI	
2436	7040		CMA		/SET DISPLAY ACTIVE FLAG.
2437	3060		DCA	ACTFLG	
2440	6031		KSF		/KEYBOARD INTERRUPT?
2441	7410		SKP		
2442	5250		JMP	ALT1+2	/YES. GO ON TO ALTMODE STRING.
2443	4420		PLCE		
2444	6042		TCF		/CLEAR TTO FLAG AND PLOT INT ENABLE.
2445	5776'		JMP	C	/NO. JUMP TO C STRING AND CONTINUE.
2446	6031	ALT1,	KSF		/ENTER FROM REQUES. WAIT FOR
2447	5246		JMP	.-1	/ALTMODE INPUT
2450	6036		KRB		
2451	1310		TAD	M375	
2452	7440		SZA		
2453	5775'		JMP	QUES	/INPUT OTHER THAN ALTMODE.
2454	4430		JMS I	CRLFP	
2455	1307		TAD	K333	/INPUT IS ALTMODE.
2456	4435		JMS I	TYPEP	/TYPE "[",
2457	1774'		TAD	PLCT	/SET COMMAND TEMPS TO 0.
2460	3305		DCA	ALTTEM	
2461	1110		TAD	M24	
2462	3306		DCA	ALTCNT	
2463	2305		ISE	ALTTEM	
2464	3705		DCA I	ALTTEM	
2465	2306		ISE	ALTCNT	
2466	5263		JMP	.-3	
2467	4773'		JMS	LISN	/LOOK FOR NEXT INPUT.
2470	4772'		JMS	QD	
2471	5771'		JMP	D	/D STRING
2472	4770'		JMS	QL	
2473	5767'		JMP	L	/L STRING
2474	4766'		JMS	QC	
2475	5776'		JMP	C	/C STRING.
2476	4765'		JMS	QT	
2477	5764'		JMP	T	/T STRING.
2500	4763'		JMS	QI	
2501	5762'		JMP	IS	/I STRING.
2502	4761'		JMS	QM	
2503	5760'		JMP	MS	/M STRING.

2631	6001	ION		
2632	7604	LAS		/WAIT TILL USER SETS
2633	7700	SMA	CLA	/SR0 TO A 1.
2634	5232	JMP	.-2	
2635	5211	JMP	MS1	
2636	1261	TAD	MSTEMP	/CONTINUE WITH M STRING.
2637	0067	AND	K100	/PEN DOWN?
2640	7640	SZA	CLA	
2641	5247	JMP	.-6	
2642	1055	TAD	PENFLG	/NO. PEN UP.
2643	7650	SNA	CLA	/PEN ALREADY UP?
2644	5253	JMP	MS2	/YES.
2645	4447	JMS I	DPUP	/NO. PUT PEN UP.
2646	5253	JMP	MS2	/GO TO NEXT PART.
2647	1055	TAD	PENFLG	/PEN DOWN.
2650	7640	SZA	CLA	/PEN ALREADY DOWN?
2651	5253	JMP	MS2	/YES.
2652	4450	JMS I	DPDP	/NO. PUT PEN DOWN.
2653	1261	TAD	MSTEMP	/GET SR.
2654	0102	AND	K77	/MASK DR.
2655	4421	PLSF		/WAIT FOR PLOT FLAG.
2656	5255	JMP	.-1	
2657	4426	CFLR		/CLEAR FLAG, LOAD DR, SET FLAG.
2660	5211	JMP	MS1	/CHECK SR AGAIN.
2661	0000	MSTEMP,	0	/SR TEMP.
2662	0177	K177,	177	
2663	5063	MSG23P,	MSG23	

/D STRING. CLEARS LOOP FLAG.

2664	7200	D,	CLA		
2665	3054		DCA	LOPFLG	/CLEAR LOOP FLAG
2666	5774		JMP	DL	/D-L STRING.

/L STRING. SETS LOOP FLAG.

2667	7240	L,	CLA	CMA	
2670	3054		DCA	LOPFLG	/SET LOOP FLAG
2671	5774		JMP	DL	/D-L STRING.

/D-L STRING. DETERMINES OPERANDS FOR D AND L COMMANDS. IF LOOP /FLAG IS SET (COMMAND L), LOOPS ON EXECUTION.

2774	3000			
2775	3400			
2776	3443			
2777	4200			
	3000	*3000		
3000	4445	DL,	JMS I	CPUP
3001	3055		DCA	PENFLG
3002	4777		JMS	LISN

3071	5241	JMP	DLSEML	/NO. GET NEXT PAT-LIN SET.
3072	1054	DLSEMC, TAD	LOPFLG	/YES. LOOP SET?
3073	7640	SZA CLA		
3074	5232	JMP	DLSEM+1	/YES. DO IT ALL AGAIN
3075	5765	JMP	REQUES	/NO. ASK FOR NEXT COMMAND.
3076	0000	DLTEM1, 0		
3077	0000	DLCNT, 0		
3100	0000	DL1, 0		
3101	0000	DL2, 0		

/C STRING: CONTINUES PREVIOUS COMMAND IF NOT COMPLETED.
 /IF PREVIOUS COMMAND COMPLETED, RESULTS IN SYNTAX ERROR.
 /IF COMMAND IS "ECL" SET LOOP FLAG AND CONTINUE; IF "ECD"
 /CLEAR LOOP FLAG; IF "EC" LEAVE LOOP FLAG IN PRESENT STATUS.

//
 //

3102	7200	G,	CLA		
3103	1060	TAD	ACTFLG	/PATTERN INTERRUPTED?	
3104	7650	SNA CLA			
3105	5774	JMP	QUES	/NO. SYNTAX ERROR. NOTHING TO CONTINUE.	
3106	4777	JMS	LISN	/LOOK FOR NEW CHARACTER.	
3107	4776	JMS	QCR	/CR?	
3110	5326	JMP	COUT	/YES. CONTINUE AS BEFORE.	
3111	4762	JMS	QL	/L"?	
3112	5316	JMP	,+4	/YES.	
3113	4761	JMS	QD	/D"?	
3114	5320	JMP	,+4	/YES.	
3115	5774	JMP	QUES	/SYNTAX ERROR. "ECX"	
3116	7240	CLA CMA		/L". SET LOOP FLAG.	
3117	7410	SKP			
3120	7200	CLA		/D". CLEAR LOOP FLAG.	
3121	3054	DCA	LOPFLG		
3122	4777	JMS	LISN	/LOOK FOR CR.	
3123	4776	JMS	QCR	/CR?	
3124	7410	SKP		/YES.	
3125	5774	JMP	QUES	/NO. SYNTAX ERROR. "ECLX" OR "ECDX"	
3126	4445	JMS I	CPUP	/YES. SET PLOT FLAG.	
3127	1055	TAD	PENFLG	/WAS PEN UP OR PEN DOWN?	
3130	7640	SZA CLA			
3131	5334	JMP	,+3		
3132	4447	JMS I	DPUP	/PEN UP.	
3133	7410	SKP			
3134	4450	JMS I	DPOP	/PEN DOWN.	
3135	1052	TAD	SAVEL	/RESTORE LINK AND AC.	
3136	7104	CLL RAL			
3137	1051	TAD	SAVEAC		
3140	6001	ION		/SET INTERRUPT.	
3141	5400	JMP I	0000	/EXIT.	

/T STRING. KILLS MONITOR AND GOES TO CONTROL TEST.

3215	5776'	JMP	QUES	/YES.
3216	7240	CLA	CMA	/NO. SET PATTERN FLAG
3217	3346	DCA	OPFLG	
3220	3347	DCA	FINFLG	/CLEAR DONE FLAG.
3221	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3222	7200	CLA		
3223	3346	DCA	OPFLG	/SET NUMBER FLAG
3224	3347	DCA	FINFLG	/CLEAR DONE FLAG
3225	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3226	3743	DCA	I OP1	/STORE PAT NUM IN TEMP.
3227	7201	CLA	IAC	/SET LINE FLAG.
3230	3346	DCA	OPFLG	
3231	7240	CLA	CMA	/SET DONE FLAG.
3232	3347	DCA	FINFLG	
3233	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3234	7200	CLA		/SET NUMBER FLAG.
3235	3346	DCA	OPFLG	/
3236	3347	DCA	FINFLG	/CLEAR DONE FLAG.
3237	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3240	3744	DCA	I OP2	/PUT LINE NUM IN TEMP.
3241	7240	CLA	CMA	
3242	3347	DCA	FINFLG	/SET DONE FLAG.
3243	7240	CLA	CMA	
3244	3346	DCA	OPFLG	/REMOVE NUMBER FLAG.
3245	4775'	JMS	LISN	/GET NEXT INPUT.
3246	5255	JMP	OPQ1	/LOOK FOR CR OR ";"
3247	0000	OPQ,	0	
3250	4775'	JMS	LISN	
3251	4774'	JMS	QP	
3252	5302	JMP	P	/"P"
3253	4773'	JMS	QL	
3254	5307	JMP	YL	/"L"
3255	4772'	OPQ1,	JMS QSEM	
3256	5314	JMP	SEM	/";"
3257	4771'	JMS	QCR	
3260	5321	JMP	CR	/CR.
3261	4770'	JMS	QNUM	/IS INPUT NUMBER? (OCTAL)
3262	7410	SKP		
3263	5776'	JMP	QUES	/NO. SYNTAX ERROR.
3264	7104	CLL	RAL	/YES. SAVE NUMBER
3265	7006	RTL		
3266	3350	DCA	OPTEM	
3267	4775'	JMS	LISN	/GET ANOTHER INPUT.
3270	4770'	JMS	QNUM	/IS INPUT NUMBER? (OCTAL)
3271	7410	SKP		
3272	5776'	JMP	QUES	/NO. SYNTAX ERROR.
3273	1350	TAD	OPTEM	/YES. PUT NUMBER IN PROPER FORM.
3274	3350	DCA	OPTEM	
3275	1346	TAD	OPFLG	/DID WE WANT NUMBERS?
3276	7640	SZA	CLA	
3277	5776'	JMP	QUES	/NO. SYNTAX ERROR
3300	1350	TAD	OPTEM	/YES. EXIT.
3301	5647	JMP	I OPQ	
3302	7200	P,	CLA	/"P"

3367 5637
 3370 3473
 3371 3443
 3372 3450
 3373 3412
 3374 3455
 3375 4200
 3376 3400
 3377 5664
 3400

*3400

3400 4430
 3401 1204
 3402 4435
 3403 5777
 3404 0277

QUES, JMS I CRLFP
 TAD K277
 JMS I TYPEP
 JMP REQU

/TYPE "?"

/EXIT TO REQUEST.

K277, 277
 /CHARACTER DETECTION SUBROUTINES.

3405 0000
 3406 4262
 3407 7474
 3410 2205
 3411 5605
 3412 0000
 3413 4262
 3414 7464
 3415 2212
 3416 5612
 3417 0000
 3420 4262
 3421 7475
 3422 2217
 3423 5617
 3424 0000
 3425 4262
 3426 7454
 3427 2224
 3430 5624
 3431 0000
 3432 4262
 3433 7467
 3434 2231
 3435 5631
 3436 0000
 3437 4262
 3440 7463
 3441 2236
 3442 5636
 3443 0000
 3444 4262
 3445 7563
 3446 2243
 3447 5643
 3450 0000
 3451 4262
 3452 7505

QD, 0 /"D"?
 JMS Q
 -304
 ISZ QD
 JMP I QD
 QL, 0 /"L"?
 JMS Q
 -314
 ISZ QL
 JMP I QL
 QC, 0 /"C"?
 JMS Q
 -303
 ISZ QC
 JMP I QC
 QT, 0 /"T"?
 JMS Q
 -324
 ISZ QT
 JMP I QT
 QI, 0 /"I"?
 JMS Q
 -311
 ISZ QI
 JMP I QI
 QM, 0 /"M"?
 JMS Q
 -315
 ISZ QM
 JMP I QM
 QCR, 0 /"CR"?
 JMS Q
 -215
 ISZ QCR
 JMP I QCR
 QSEM, 0 /"I"?
 JMS Q
 -273

3520	6001	ION		/SET INTERRUPT
3521	7200	CLA		
3522	1335	TAD	STRTL	
3523	3776	DCA	LINUM	
3524	1336	TAD	STRTP	
3525	4775	JMS	LINCNT	/GET START POINTER.
3526	3774	DCA	START	/SAVE.
3527	1337	TAD	STPL	
3530	3776	DCA	LINUM	
3531	1340	TAD	STPP	
3532	4775	JMS	LINCNT	/GET STOP POINTER.
3533	3773	DCA	STOP	/SAVE.
3534	3772	JMP	LGENL1	
3535	0000	STRTL,	0	
3536	0000	STRTP,	0	
3537	0000	STPL,	0	
3540	0000	STPP,	0	
/				
/				
/				
/				
3572	3600			
3573	3731			
3574	3730			
3575	4000			
3576	4050			
3577	2423			
	3600	*3600		
3600	7300	LGENL1,	CLA CLL	
3601	1730	TAD I	START	/IS PRESENT LID STANDARD OR SPECIAL.
3602	7006	RTL		
3603	7420	SNL		
3604	5243	JMP	STD	/STANDARD.
3605	7500	SMA		
3606	5243	JMP	STD	/STANDARD.
3607	7012	SPEC,	RTR	/SPECIAL (3XXY).
3610	0076	AND	K7	/EXTRACT Y, NEGATE AND SAVE.
3611	7041	CIA		
3612	3332	DCA	SPTALH	
3613	1730	TAD I	START	/EXTRACT XX, COMPUTE REPETITION
3614	7010	RAR		/CONSTANT POINTER, GET REPETITION
3615	7012	RTR		
3616	0102	AND	K77	
3617	1777	TAD	REPK	/CONSTANT, NEGATE, AND PUT IN
3620	3333	DCA	SPT1	/COUNTER.
3621	1733	TAD I	SPT1	
3622	7041	CIA		
3623	3334	DCA	SPREP	
3624	1330	SPECL1,	TAD	/SET UP SPECIAL LID STARTING POINTER.
3625	3335	DCA	SP1P	
3626	1332	TAD	SPTALH	/SET UP LID COUNTER.
3627	3336	DCA	SPTAL	

```

3715 0343          AND      K37          /CONSTANT POINTER AND SAVE.
3716 1774          TAD      DKAYS
3717 3342          DCA      DKP
3720 1742          TAD I    DKP          /PUT DIRECTION CONSTANT IN AC.
3721 4421          XLATEL, PLSF          /WAIT FOR PLOTTER FLAG.
3722 5321          JMP      .-1
3723 4426          CFLR
3724 2341          ISZ      LENGTH      /DRAW INCREMENT.
3725 5321          JMP      XLATEL      /COMPLETE LID DRAWN?
3726 7300          CLA CLL              /NO. DRAW ANOTHER INCREMENT.
3727 5666          JMP I    XLATE      /YES.
3730 0000          START, 0            /EXIT.
3731 0000          STOP, 0
3732 0000          SPTALH, 0
3733 0000          SPT1, 0
3734 0000          SPREP, 0
3735 0000          SP1P, 0
3736 0000          SPTAL, 0
3737 0000          LID, 0
3740 0000          XLGP, 0
3741 0000          LENGTH, 0
3742 0000          DKP, 0
3743 0037          K37, 37
3744 3740          K3740, 3740

```

/LINE COUNT ROUTINE: ENTERED BY JMS LINCNT WITH PATTERN NUMBER
 /IN AC AND LINE NUMBER IN "LINUM"
 /1. COMPUTES ADDRESS OR POINTER OF SPECIFIC LINE WITHIN
 /PATTERN THROUGH USE OF PATTERN NUMBER AND LINE NUMBER.
 /2. THE GENERAL METHOD IS TO TREAT SPECIAL LIDS (3XXY) WITH
 /THE NEXT Y LIDS AS ONE LINE AND ALL STAND ALONE STANDARD LIDS
 /AS ONE LINE.
 /3. EXITS WITH LINE POINTER IN AC.

```

3774 5404
3775 5457
3776 3516
3777 5622
      4000
4000 0000          *4000
4001 1777          LINCNT, 0
4002 3246          TAD      PATS          /COMPUTE PATTERN POINTER
4003 1646          DCA      PATPT        /AND SAVE.
4004 3246          TAD I    PATPT
4005 1246          DCA      PATPT
4006 3247          TAD      PATPT
4007 1250          DCA      LIDPT        /SET LID POINTER TO PATTERN
4010 7041          TAD      LINUM        /POINTER.
4011 3251          CIA      LINTAL      /SET LINE TALLY EQUAL TO
4012 3252          DCA      ADDCNT      /-LINE NUMBER.
4013 1647          DCA      LIDPT
4014 7106          LINL1, TAD I    LIDPT /SET ADDRESS COUNTER TO 0.
4015 7420          CLL RTL              /GET LID.
          SNL                          /IS LID SPECIAL OR STD?

```

```

4076 5275      JMP      .-1
4077 4425      PLPD
4100 3055      DCA      PENFLG      /PEN DOWN,
4101 5673      JMP I    DPDS      /SET PEN FLAG.

```

```

/
/CONTROL TEST TYPE, 600-700 SERIES.
CPU8, 0
4102 0000      CLA
4103 7200      TAD      K31      /PEN UP.
4104 1077      CFLR
4105 4426      CLA
4106 7200      JMP I    CPU8
4107 5702      CPD8, 0
4110 0000      CLA
4111 7200      TAD      K32      /PEN DOWN
4112 1100      CFLR
4113 4426      CLA
4114 7200      JMP I    CPD8
4115 5710

```

```

/DISPLAY TEST TYPE, 600-700 SERIES.
DPUS, 0
4116 0000      CLA
4117 7200      TAD      K31
4120 1077      PLSF
4121 4421      JMP      .-1
4122 5321      CFLR      /PEN UP.
4123 4426      CLA
4124 7200      DCA      PENFLG      /CLEAR PEN FLAG.
4125 3055      JMP I    DPUS
4126 5716      DPD8, 0
4127 0000      CLA
4130 7200      TAD      K32
4131 1100      PLSF
4132 4421      JMP      .-1
4133 5332      CFLR      /PEN DOWN.
4134 4426      CLA CMA
4135 7240      DCA      PENFLG      /SET PEN FLAG.
4136 3055      JMP I    DPD8
4137 5727

```

```

/DECREASE AND SKIP IF ZERO SUBROUTINE. (AC AND LINK ARE PRESERVED.)
/ENTERED WITH:
/      JMS I    DSZP      /LINK DSZP, DSZ
/      ABCD      /COUNTER POINTER.
/      NEXT INST.
/DECREASES CONTENTS OF ABCD BY 1 AND SKIPS NEXT INST IF 0.

```

```

DSZ, 0
4140 0000      DCA      DSZT1      /SAVE AC.
4141 3354      TAD I    DSZ      /GET COUNTER POINTER.
4142 1740      DCA      DSZT2
4143 3355      ISZ      DSZ      /UPDATE RETURN ADDRESS.
4144 2340      CLA CMA      /DECREASE COUNTER BY 1.
4145 7240      TAD I    DSZT2
4146 1755      SNA
4147 7450      /COUNTER=0?

```

```

4217 0000 READN, 0
4220 4200      JMS     LISN
4221 0223      AND     K17
4222 5617      JMP I   READN
4223 0017      K17,    17
          /RING TTY BELL SUBROUTINE.

```

```

4224 0000 BELL, 0
4225 7200      CLA
4226 1231      TAD     K207
4227 4435      JMS I   TYPEP
4230 5624      JMP I   BELL
4231 0207      K207,   207

```

/TYPE PACKED MESSAGE SUBROUTINE. ENTERED WITH POINTER IN AC.

```

4232 0000 MSGN, 0
4233 3270      DCA     MSGTEM
4234 7240      CLA CMA
4235 1270      TAD     MSGTEM
4236 3017      DCA     AUTO7
4237 1107      MSGNL1, TAD     M2
4240 3272      DCA     MSGCNT
4241 1417      TAD I    AUTO7
4242 7004      RAL
4243 3270      DCA     MSGTEM
4244 7620      MSGNL2, SNL CLA
4245 5250      JMP     .+3
4246 1101      TAD     K200
4247 5251      JMP     .+2
4250 1273      TAD     K300
4251 3271      DCA     MSGK
4252 1270      TAD     MSGTEM
4253 7006      RTL
4254 7006      RTL
4255 7006      RTL

```

```

4256 3270      DCA     MSGTEM
4257 1270      TAD     MSGTEM
4260 0102      AND     K77
4261 7450      SNA
4262 5632      JMP I   MSGN
4263 1271      TAD     MSGK
4264 4435      JMS I   TYPEP
4265 2272      ISZ     MSGCNT
4266 5244      JMP     MSGNL2
4267 5237      JMP     MSGNL1
4270 0000 MSGTEM, 0
4271 0000 MSGK, 0
4272 0000 MSGCNT, 0
4273 0300 K300, 300

```

/DELAY N MILLISEC ROUTINE. ENTERED WITH N IN AC.

4426	6505	6505		
4427	5625	JMP I	SPLPD	
4430	7402	HLT		/ILLEGAL SKIP.
4431	0000	SCFLR,	0	/CFLR.
4432	6506	6506		
4433	5631	JMP I	SCFLR	
4434	7402	HLT		/ILLEGAL SKIP.
4435	0000	SPLSE,	0	/PLSE
4436	6507	6507		
4437	5635	JMP I	SPLSE	
4440	7402	HLT		/ILLEGAL SKIP.

/MESSAGES.

4600	3031	*4600	
4601	4070	MSG1,	TEXT "XY 8-E DIAGNOSTIC PROGRAM"
4602	5505		
4603	4004		
4604	1101		
4605	0716		
4606	1723		
4607	2411		
4610	0340		
4611	2022		
4612	1707		
4613	2201		
4614	1500		
4615	2014	MSG2,	TEXT "PLOTTER SERIES (1-3): *"
4616	1724		
4617	2405		
4620	2240		
4621	2305		
4622	2211		
4623	0523		
4624	4050		
4625	6155		
4626	6351		
4627	7240		
4630	5200		
4631	2014	MSG3,	TEXT "PLOTTER TYPE (1-4): *"
4632	1724		
4633	2405		
4634	2240		
4635	2431		
4636	2005		
4637	4050		
4640	6155		
4641	6451		
4642	7240		
4643	5200		
4644	2605	MSG4,	TEXT "VECTOR TYPE (1-2): *"
4645	0324		
4646	1722		

4736	1116	MSG12, TEXT	"INCREMENT SIZE: "
4737	0522		
4740	0515		
4741	0516		
4742	2440		
4743	2311		
4744	3205		
4745	7240		
4746	0000		
4747	0317	MSG13, TEXT	"CORRECT? (Y OR N) *"
4750	2222		
4751	0503		
4752	2477		
4753	4050		
4754	3140		
4755	1722		
4756	4016		
4757	5140		
4760	5200		
4761	0317	MSG14, TEXT	"CONTROL TEST."
4762	1624		
4763	2217		
4764	1440		
4765	2405		
4766	2524		
4767	5600		
4770	4040	MSG15, TEXT	" ERROR PC DR"
4771	0522		
4772	2217		
4773	2240		
4774	4040		
4775	4020		
4776	0340		
4777	4040		
5000	0422		
5001	0000		
5002	0317	MSG17, TEXT	"CONTROL TEST COMPLETE."
5003	1624		
5004	2217		
5005	1440		
5006	2405		
5007	2524		
5010	4003		
5011	1715		
5012	2014		
5013	0524		
5014	0556		
5015	0000		
5016	0411	MSG20, TEXT	"DISPLAY TEST MONITOR."
5017	2520		
5020	1401		
5021	3140		
5022	2405		
5023	2524		
5024	4015		

```

5114 0000 M16P2, 0 /PC FILL IN
5115 0000 M16P3, 0
5116 4040 4040
5117 0000 M16P4, 0 /DIRECTION REG. FILL IN.
5120 0000 0000 /END MSG CODE.

```

/VERIFICATION MESSAGE INSERTS.
PSM1, TEXT "500 OR HOUSTON DP10/EDP10"

```

5121 6560
5122 6040
5123 1722
5124 4010
5125 1725
5126 2624
5127 1716
5130 4004
5131 2061
5132 6057
5133 0504
5134 2061
5135 6000
5136 6660 PSM2, TEXT "600"
5137 6000
5140 6760 PSM3, TEXT "700"
5141 6000
5142 6161 PTM1, TEXT "11 INCH DRUM OR HOUSTON DP10/EDP10"
5143 4011
5144 1603
5145 1040
5146 0422
5147 2515
5150 4017
5151 2240
5152 1017
5153 2523
5154 2417
5155 1640
5156 0420
5157 6160
5160 5705
5161 0420
5162 6160
5163 0000
5164 6270 PTM2, TEXT "28.55 INCH DRUM"
5165 5665
5166 6740
5167 1116
5170 0310
5171 4004
5172 2225
5173 1500
5174 6361 PTM3, TEXT "31 BY 54 INCH FLATBED"
5175 4002
5176 3140
5177 6564
5200 4011

```

5264	0036	30	/31 X 54 INCH FLATBED.
5265	0065	53	
5266	0065	53	/54 X 72 INCH FLATBED.
5267	0107	71	
		OCTAL	
5270	5270	INSTP.	/SCALE FACTOR INSTS FOR INIT.
5271	7104	CLL RAL	/.00125 INCH SCALE FACTOR (2.0)
5272	7000	NOP	
5273	7000	NOP	/.0025 INCH SCALE FACTOR (1.0)
5274	7000	NOP	
5275	7110	CLL RAR	/.005 INCH SCALE FACTOR (0.5)
5276	7000	NOP	
5277	7110	CLL RAR	/.01 INCH SCALE FACTOR (0.25)
5300	7110	CLL RAR	

/INIT REPOSE MSG POINTERS

5301	5121	PSM1P,	PSM1
5302	5136	PSM2P,	PSM2
5303	5140	PSM3P,	PSM3
5304	5142	PTM1P,	PTM1
5305	5164	PTM2P,	PTM2
5306	5174	PTM3P,	PTM3
5307	5207	PTM4P,	PTM4
5310	5222	VTM1P,	VTM1
5311	5223	VTM2P,	VTM2
5312	5225	ITM1P,	ITM1
5313	5231	ITM2P,	ITM2
5314	5237	ITM3P,	ITM3
5315	5245	ITM4P,	ITM4
5316	5252	ITM5P,	ITM5

/ACTIVE PATTERN POINTERS. (FILLED IN BY INIT)

5317	5317	PATS.	.	
5320	0000		0	/PAT1
5321	0000		0	/PAT2
5322	0000		0	/PAT3
5323	0000		0	/PAT4
5324	0000		0	/PAT5
5325	0000		0	/PAT6
5326	6721		PAT07	/PAT07

/24 VECTOR PATTERN POINTERS.

5327	5327	PATS24,	.	
5330	6271		PAT124	/PAT1
5331	6341		PAT224	/PAT2
5332	6407		PAT324	/PAT3
5333	6447		PAT424	/PAT4
5334	6556		PAT524	/PAT5
5335	6647		PAT624	/PAT6

/8 VECTOR PATTERN POINTERS.

5402 4116 DPU8P, DPU8 /DISPLAY PEN UP
 5403 4127 DPD8P, DPD8 /DISPLAY PEN DOWN

/DIRECTION CONSTANTS. THOSE DEFINED AS ZERO ARE REDEFINED BY
 /PLOTTER SERIES CONNECTED. ADDRESS SENSITIVE RELATIVE TO
 /DKAYS".

5404	5404	DKAYS,	.	/COORDINATE LID CODE	
5405	0000	NK,	0	/+Y	01
5406	0000	NEK,	0	/+Y +X	02
5407	0000	EK,	0	/	03
5410	0000	SEK,	0	/-Y +X	04
5411	0000	SK,	0	/-Y	05
5412	0000	SWK,	0	/-Y -X	06
5413	0000	WK,	0	/	07
5414	0000	NWK,	0	/+Y -X	10
5415	0072	NNEK,	72	/+Y +X/2	11
5416	0070	ENEK,	70	/+Y/2 +X	12
5417	0074	ESEK,	74	/-Y/2 +X	13
5420	0076	SSEK,	76	/-Y +X/2	14
5421	0077	SSWK,	77	/-Y -X/2	15
5422	0075	WSWK,	75	/-Y/2 -X	16
5423	0071	WNWK,	71	/+Y/2 -X	17
5424	0073	NNWK,	73	/+Y -X/2	20
5425	0050	LNK,	50	/+Y/2	21
5426	0051	LNEK,	51	/+Y/2 +X/2	22
5427	0052	LEK,	52	/	23
5430	0053	LSEK,	53	/-Y/2 +X/2	24
5431	0054	LSK,	54	/-Y/2	25
5432	0055	LSWK,	55	/-Y/2 -X/2	26
5433	0056	LWK,	56	/	27
5434	0057	LNWK,	57	/+Y/2 -X/2	30

/8 VECTOR DIRECTION CONSTANTS DEFINITIONS. ADDRESS SENSITIVE
 /RELATIVE TO "DK500" AND "DK678".
 /500 SERIES.

5435	5435	DK500,	.
5436	0020	NK5,	20
5437	0030	NEK5,	30
5440	0010	EK5,	10
5441	0050	SEK5,	50
5442	0040	SK5,	40
5443	0044	SWK5,	44
5444	0004	WK5,	04
5445	0024	NWK5,	24

/600-700 SERIES.

5446	5446	DK678,	.
5447	0010	NK8,	10
5450	0011	NEK8,	11
5451	0012	EK8,	12

/LENGTH CONSTANTS FOR BASIC .0025 INCH INCREMENT SIZE. (IN DECIMAL.)

5530	5530	INCH.	DECIMAL	/GROUP 1.
5531	0050		40	/00
5532	0144		100	/01
5533	0310		200	/02
5534	0620		400	/03
5535	1130		600	/04
5536	1750		1000	/05
5537	2260		1200	/06
5540	3100		1600	/07
				/GROUP2. (.707 X GROUP1)
5541	0034		28	/10
5542	0107		71	/11
5543	0215		141	/12
5544	0433		283	/13
5545	0650		424	/14
5546	1303		707	/15
5547	1520		848	/16
5550	2153		1131	/17
				/GROUP3. (.447 X GROUP4)
5551	0044		36	/20
5552	0131		89	/21
5553	0263		179	/22
5554	0546		358	/23
5555	1030		536	/24
5556	1576		894	/25
5557	2061		1073	/26
5560	2626		1430	/27
				/GROUP4 (2 X GROUP1)
5561	0120		80	/30
5562	0310		200	/31
5563	0620		400	/32
5564	1440		800	/33
5565	2260		1200	/34
5566	3720		2000	/35
5567	4540		2400	/36
5570	6200		3200	/37
				/GROUP5 (.707 X GROUP4)
5571	0071		57	/40
5572	0215		141	/41
5573	0433		283	/42
5574	1066		566	/43
5575	1520		848	/44
5576	2606		1414	/45
5577	3241		1697	/46
5600	4526		2262	/47

OCTAL

/LENGTH CONSTANTS FOR 0.1 MM INCREMENT SIZE. (IN DECIMAL)

5653	0000	0
5654	0000	0
5655	0000	0
5656	0000	0
5657	0000	0
5660	0000	0
5661	0000	0
5662	0000	0
5663	0000	0

/PATTERN-LINE COMMAND TEMPORARIES.

5664	5664	PLCT,	.
5665	0000	PCT,	0
5666	0000		0
5667	0000		0
5670	0000		0
5671	0000		0
5672	0000		0
5673	0000		0
5674	0000		0
5675	0000		0
5676	0000		0
5677	0000	LCT,	0
5700	0000		0
5701	0000		0
5702	0000		0
5703	0000		0
5704	0000		0
5705	0000		0
5706	0000		0
5707	0000		0
5710	0000		0

/PATTERN LISTS.

/1. THERE ARE TWO MAJOR GROUPS OF PATTERNS: 8 VECTOR PATTERNS (PAT 18, 28, 38, 48, 58, AND 68) AND 24 VECTOR PATTERNS (PAT 124, 224, 324, 424, 524, AND 624). PAT 07 /IS A LIST OF DASH AND SHUFFLE VECTORS WHICH ARE CALLED ON A LINE TO LINE BASIS /BY THE USER. THE VECTORS INCLUDED IN PAT07 REPRESENT ALL 16 DIRECTIONS.

/2. THE PATTERN LISTS CONSIST OF LINE IDENTIFIERS (LID) WHICH ARE /IN EFFECT PSEUDO INSTRUCTIONS WHICH DESCRIBE PEN UP OR DOWN, LINE LENGTH, AND /DIRECTION. THERE IS ALSO A SPECIAL GROUP OF LID'S. LID'S ARE INTERPRETED BY THE /"LGEN" ROUTINE. THE FORMATS ARE AS FOLLOWS:

/STANDARD LID BIT ASSIGNMENTS

/0: PEN UP WHEN CLEAR, PEN DOWN WHEN SET.

/1-6: LENGTH CODE (REFERS TO SPECIFIED LENGTH CONSTANT).

/7-11: DIRECTION CODE (REFERS TO DIRECTION CONSTANT).

/SPECIAL LID (3XXY) FORMAT.

/3: INDICATES SPECIAL LID.

/XX: REFERS TO REPETITION CONSTANT.

/Y: INDICATES HOW MANY STANDARD LID'S FOLLOWING THE SPECIAL

/LID ARE TO BE INCLUDED IN THIS LINE.

/IN ALL CASES +Y=PEN LEFT; -Y=PEN RIGHT; +X=DRUM DOWN; -X=DRUM UP.

/PATTERN 28-DASH FILL IN .25 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR	
5761	3011	PAT28, 3011	/01				
5762	0101	0101	/	0	02	01	
5763	0047	0047	/02	0	01	07	
5764	3022	3022	/03				
5765	4045	4045	/	1	01	05	
5766	0045	0045	/	0	01	05	
5767	0042	0042	/04	0	01	02	
5770	3062	3062	/05				
5771	4047	4047	/	1	01	07	
5772	0047	0047	/	0	01	07	
5773	0044	0044	/06	0	01	04	
5774	3022	3022	/07				
5775	4041	4041	/	1	01	01	
5776	0041	0041	/	0	01	01	
5777	0046	0046	/10	0	01	06	
6000	3062	3062	/11				
6001	4043	4043	/	1	01	03	
6002	0043	0043	/	0	01	03	
6003	3062	3062	/12				
6004	4047	4047	/	1	01	07	
6005	0047	0047	/	0	01	07	
6006	0042	0042	/13	0	01	02	
6007	3022	3022	/14				
6010	4045	4045	/	1	01	05	
6011	0045	0045	/	0	01	05	
6012	0050	0050	/15	0	01	10	
6013	3062	3062	/16				
6014	4043	4043	/	1	01	03	
6015	0043	0043	/	0	01	03	
6016	0046	0046	/17	0	01	06	
6017	3022	3022	/20				
6020	4041	4041	/	1	01	01	
6021	0041	0041	/	0	01	01	
6022	0043	0043	/21	0	01	03	
6023	3011	L28, 3011	/22				PAT02 ORIGIN
6024	0105	0105	/	0	02	05	
6025	3051	3051	/23				PAT03 ORIGIN
6026	0147	0147	/	0	03	07	

/PATTERN 38-SHUFFLE .5 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR	
6027	3011	PAT38, 3011	/01				
6030	0101	0101	/	0	02	01	
6031	0103	0103	/02	0	02	03	
6032	3014	3014	/03				
6033	4105	4105	/	1	02	05	
6034	0105	0105	/	0	02	05	
6035	4101	4101	/	1	02	01	
6036	0105	0105	/	0	02	05	
6037	0110	0110	/04	0	02	10	
6040	3054	3054	/05				
6041	4103	4103	/	1	02	03	

6124	4341	4341	/31	1	07	01
6125	4347	4347	/32	1	07	07
6126	3111	3111	/33			
6127	0144	0144	/	0	03	04
6130	0341	0341	/34	0	07	01
6131	4344	4344	/35	1	07	04
6132	4346	4346	/36	1	07	06
6133	4350	4350	/37	1	07	10
6134	4342	4342	/40	1	07	02
6135	0345	0345	/41	0	07	05
6136	3051	3051	/42			
6137	0107	0107	/	0	02	07
			/			
			/			
			/			

4 INCH OBLIQUE SQ.

PAT04 ORIGIN
PAT05 ORIGIN

/PATTERN 58 - JOINED OCTAGONS

			/LINE	PEN	LGTH	DIR
6140	4141	PAT58, 4141	/01	1	03	01
6141	4041	4041	/02	1	01	01
6142	4642	4642	/03	1	15	02
6143	4243	4243	/04	1	05	03
6144	4644	4644	/05	1	15	04
6145	4245	4245	/06	1	05	05
6146	4646	4646	/07	1	15	06
6147	4247	4247	/10	1	05	07
6150	4650	4650	/11	1	15	10
6151	4241	4241	/12	1	05	01
6152	4650	4650	/13	1	15	10
6153	4247	4247	/14	1	05	07
6154	4646	4646	/15	1	15	06
6155	4245	4245	/16	1	05	05
6156	4644	4644	/17	1	15	04
6157	4243	4243	/20	1	05	03
6160	4642	4642	/21	1	15	02
6161	0241	0241	/22	0	05	01
6162	4344	4344	/23	1	07	04
6163	4044	4044	/24	1	01	04
6164	0642	0642	/25	0	15	02
6165	4350	4350	/26	1	07	10
6166	4050	4050	/27	1	01	10
6167	0243	0243	/30	0	05	03
6170	0544	0544	/31	0	13	04
6171	0444	0444	/32	0	11	04
6172	4346	4346	/33	1	07	06
6173	4046	4046	/34	1	01	06
6174	0450	0450	/35	0	11	10
6175	0550	0550	/36	0	13	10
6176	0241	0241	/37	0	05	01
6177	4346	4346	/40	1	07	06
6200	4046	4046	/41	1	01	06
6201	0650	0650	/42	0	15	10
6202	4342	4342	/43	1	07	02
6203	4042	4042	/44	1	01	02

START TOP OCTAGON

BOTTOM OCTAGON

TOP DIAGS.

LOW DIAGS.

6266	0143	L68,	0143	/17	0	03	03	PAT06 ORIGIN.
6267	3051		3051	/20				
6270	0107		0107	/	0	02	07	PAT01 ORIGIN.
				/				

/24 VECTOR PATTERNS
/PATTERN 124-PERIMETER OUTLINE.

/LINE# PEN LGTH DIR

6271	3011	PAT124, 3011	/01				ORIENT
6272	0141	0141	/	0	03	01	
6273	3011	3011	/02				
6274	0145	0145	/	0	03	05	
6275	3011	3011	/03				
6276	0145	0145	/	0	03	05	
6277	0101	0101	/04	0	02	01	
6300	3011	3011	/05				
6301	0141	0141	/	0	03	01	
6302	3051	3051	/06				
6303	0143	0143	/	0	03	03	
6304	0107	0107	/07	0	02	07	
6305	3051	3051	/10				PERIMETER
6306	4147	4147	/	1	03	07	
6307	3011	3011	/11				
6310	4145	4145	/	1	03	05	
6311	3051	3051	/12				
6312	4143	4143	/	1	03	03	
6313	3011	3011	/13				
6314	4141	4141	/	1	03	01	
6315	3051	3051	/14				
6316	0107	0107	/	0	02	07	
6317	3011	3011	/15				
6320	0105	0105	/	0	02	05	
6321	4041	4041	/16	1	01	01	CROSS HAIRS
6322	4105	4105	/17	1	02	05	
6323	4041	4041	/20	1	01	01	
6324	4043	4043	/21	1	01	03	
6325	4107	4107	/22	1	02	07	
6326	4043	4043	/23	1	01	03	
6327	0041	0041	/24	0	01	01	
6330	4404	4404	/25	1	10	04	
6331	0410	0410	/26	0	10	10	
6332	4406	4406	/27	1	10	06	
6333	0402	0402	/30	0	10	02	
6334	0045	0045	/31	0	01	05	
6335	3051	L124, 3051	/32				
6336	0107	0107	/	0	02	07	PAT01 ORIGIN
6337	3051	3051	/33				
6340	0143	0143	/	0	03	03	PAT02 ORIGIN
			/				

6422	0103	0103	/	0	02	03
6423	4107	4107	/	1	02	07
6424	0103	0103	/	0	02	03
6425	0106	0106	/06	0	02	06
6426	3014	3014	/07			
6427	4101	4101	/	1	02	01
6430	0101	0101	/	0	02	01
6431	4105	4105	/	1	02	05
6432	0101	0101	/	0	02	01
6433	0104	0104	/10	0	02	04
6434	3054	3054	/11			
6435	4107	4107	/	1	02	07
6436	0107	0107	/	0	02	07
6437	4103	4103	/	1	02	03
6440	0107	0107	/	0	02	07
6441	0101	0101	/12	0	02	01
6442	3011	3011	/13			
6443	0105	0105	/	0	02	05
6444	3052	3052	/14			
6445	0103	0103	/	0	02	03
6446	0043	0043	/	0	01	03

L324,

PAT03 ORIGIN

PAT04 ORIGIN

/PATTERN 424 - CONCENTRIC AND OBLIQUE SQUARES.

			/LINE	PEN	LGTH	DIR
6447	4341	PAT424, 4341	/01	1	07	01 N
6450	0345	0345	/02	0	07	05
6451	4351	4351	/03	1	07	11 NNE
6452	0355	0355	/04	0	07	15
6453	4342	4342	/05	1	07	02 NE
6454	0346	0346	/06	0	07	06
6455	4352	4352	/07	1	07	12 ENE
6456	0356	0356	/10	0	07	16
6457	4343	4343	/11	1	07	03 E
6460	0347	0347	/12	0	07	07
6461	4353	4353	/13	1	07	13 ESE
6462	0357	0357	/14	0	07	17
6463	4344	4344	/15	1	07	04 SE
6464	0350	0350	/16	0	07	10
6465	4354	4354	/17	1	07	14 SSE
6466	0360	0360	/20	0	07	20
6467	4345	4345	/21	1	07	05 S
6470	0341	0341	/22	0	07	01
6471	4355	4355	/23	1	07	15 SSW
6472	0351	0351	/24	0	07	11
6473	4346	4346	/25	1	07	06 SW
6474	0342	0342	/26	0	07	02
6475	4356	4356	/27	1	07	16 WSW
6476	0352	0352	/30	0	07	12
6477	4347	4347	/31	1	07	07 W
6500	0343	0343	/32	0	07	03
6501	4357	4357	/33	1	07	17 WNW
6502	0353	0353	/34	0	07	13
6503	4350	4350	/35	1	07	10 NW
6504	0344	0344	/36	0	07	04

6566	4145	4145	/11	1	03	05	
6567	4155	4155	/12	1	03	15	
6570	4146	4146	/13	1	03	06	
6571	4156	4156	/14	1	03	16	
6572	4147	4147	/15	1	03	07	
6573	4157	4157	/16	1	03	17	
6574	4150	4150	/17	1	03	10	
6575	4160	4160	/20	1	03	20	
6576	4141	4141	/21	1	03	01	LOW 16'ER
6577	4160	4160	/22	1	03	20	
6600	4150	4150	/23	1	03	10	
6601	4157	4157	/24	1	03	17	
6602	4147	4147	/25	1	03	07	
6603	4156	4156	/26	1	03	16	
6604	4146	4146	/27	1	03	06	
6605	4155	4155	/30	1	03	15	
6606	4145	4145	/31	1	03	05	
6607	4154	4154	/32	1	03	14	
6610	4144	4144	/33	1	03	04	
6611	4153	4153	/34	1	03	13	
6612	4143	4143	/35	1	03	03	
6613	4152	4152	/36	1	03	12	
6614	4142	4142	/37	1	03	02	
6615	4151	4151	/40	1	03	11	
6616	4244	4244	/41	1	05	04	TOP DIAGS.
6617	0143	0143	/42	0	03	03	
6620	0152	0152	/43	0	03	12	
6621	4350	4350	/44	1	07	10	
6622	0142	0142	/45	0	03	02	
6623	4344	4344	/46	1	07	04	
6624	0151	0151	/47	0	03	11	
6625	0141	0141	/50	0	03	01	
6626	4250	4250	/51	1	05	10	
6627	0246	0246	/52	0	05	06	
6630	0146	0146	/53	0	03	06	
6631	4246	4246	/54	1	05	06	LOW DIAGS
6632	0147	0147	/55	0	03	07	
6633	0157	0157	/56	0	03	17	
6634	4342	4342	/57	1	07	02	
6635	0150	0150	/60	0	03	10	
6636	4346	4346	/61	1	07	06	
6637	0160	0160	/62	0	03	20	
6640	0141	0141	/63	0	03	01	
6641	4242	4242	/64	1	05	02	
6642	0244	0244	/65	0	05	04	
6643	0144	0144	/66	0	03	04	
6644	0101	0101	/67	0	02	01	PAT05 ORIGIN
6645	3051	3051	/70				PAT06 ORIGIN
6646	0043	0043	/	0	01	03	

/PATTERN 624 - RAPID ARROWS

6647 0147

PAT624, 0147

/LINE	PEN	LGTH	DIR
/01	0	03	07

6726	4144	4144	/	1	03	04	
6727	3011	3011	/04				NE
6730	4142	4142	/	1	03	02	DASHES
6731	3012	3012	/05				N
6732	4101	4101	/	1	02	01	
6733	0101	0101	/	0	02	01	
6734	3012	3012	/06				NNE
6735	4111	4111	/	1	02	11	
6736	0111	0111	/	0	02	11	
6737	3012	3012	/07				NE
6740	4102	4102	/	1	02	02	
6741	0102	0102	/	0	02	02	
6742	3012	3012	/10				ENE
6743	4112	4112	/	1	02	12	
6744	0112	0112	/	0	02	12	
6745	3052	3052	/11				E
6746	4103	4103	/	1	02	03	
6747	0103	0103	/	0	02	03	
6750	3012	3012	/12				ESE
6751	4113	4113	/	1	02	13	
6752	0113	0113	/	0	02	13	
6753	3012	3012	/13				SE
6754	4104	4104	/	1	02	04	
6755	0104	0104	/	0	02	04	
6756	3012	3012	/14				SSE
6757	4114	4114	/	1	02	14	
6760	0114	0114	/	0	02	14	
6761	3012	3012	/15				S
6762	4105	4105	/	1	02	05	
6763	0105	0105	/	0	02	05	
6764	3012	3012	/16				SSW
6765	4115	4115	/	1	02	15	
6766	0115	0115	/	0	02	15	
6767	3012	3012	/17				SW
6770	4106	4106	/	1	02	06	
6771	0106	0106	/	0	02	06	
6772	3012	3012	/20				WSW
6773	4116	4116	/	1	02	16	
6774	0116	0116	/	0	02	16	
6775	3052	3052	/21				W
6776	4107	4107	/	1	02	07	
6777	0107	0107	/	0	02	07	
7000	3012	3012	/22				WNW
7001	4117	4117	/	1	02	17	
7002	0117	0117	/	0	02	17	
7003	3012	3012	/23				NW
7004	4110	4110	/	1	02	10	
7005	0110	0110	/	0	02	10	
7006	3012	3012	/24				NNW
7007	4120	4120	/	1	02	20	
7010	0120	0120	/	0	02	20	SHUFFLES
7011	3014	3014	/25				N
7012	4101	4101	/	1	02	01	

7100	3014	3014	/40				WSW
7101	4116	4116	/	1	02	16	
7102	0116	0116	/	0	02	16	
7103	4112	4112	/	1	02	12	
7104	0116	0116	/	0	02	16	
7105	3054	3054	/41				W
7106	4107	4107	/	1	02	07	
7107	0107	0107	/	0	02	07	
7110	4103	4103	/	1	02	03	
7111	0107	0107	/	0	02	07	
7112	3014	3014	/42				WNW
7113	4117	4117	/	1	02	17	
7114	0117	0117	/	0	02	17	
7115	4113	4113	/	1	02	13	
7116	0117	0117	/	0	02	17	
7117	3014	3014	/43				NW
7120	4110	4110	/	1	02	10	
7121	0110	0110	/	0	02	10	
7122	4104	4104	/	1	02	04	
7123	0110	0110	/	0	02	10	
7124	3014	3014	/44				NNW
7125	4120	4120	/	1	02	20	
7126	0120	0120	/	0	02	20	
7127	4114	4114	/	1	02	14	
7130	0120	0120	/	0	02	20	

S